Maastricht University

## Master Thesis

# Machine Learning for Pre-Radiotherapy Tooth Extraction Prediction

*Confidential:* No


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# 1 Introduction

Head and neck cancer (HNC) is the seventh most common cancer worldwide (Gormley et al., 2022). Radiotherapy (RT) is a widely used treatment for HNC, benefiting approximately 75% of patients (Gronberg et al., 2023). However, RT is known to cause both early and late toxicities, including osteoradionecrosis (ORN), a serious complication that can significantly affect the quality of life (QoL) of patients and survivors (Yang et al., 2024). To reduce the risk of developing ORN, teeth with a poor prognosis are often extracted before RT. However, recent findings, such as Buurman et al. (2023) and Karaca et al. (2024) suggest that a considerable number of these extractions may be unnecessary. Beyond their social and aesthetic implications, tooth extractions can impair chewing function, potentially affecting a patient's QoL and adherence to treatment, particularly due to complications such as weight loss (Karaca et al. 2024). Maintaining body weight is crucial for completing RT as planned and achieving optimal recovery outcomes (Buurman et al., 2023). As such, minimizing avoidable extractions is essential to prevent additional weight loss and associated risks. Yet despite the clinical relevance, there has been limited exploration into predictive approaches that could help identify which teeth truly require extraction prior to RT.

The decision to extract teeth is made before the detailed RT planning is available. As a result, predictive models must rely on a limited set of clinical features, most of which are categorical in nature. While some machine learning (ML) methods can handle categorical inputs directly, others, depend on one hot encoding (OHE). This transformation can significantly expand feature dimensionality, particularly with high-cardinality variables, raising concerns about overfitting and computational inefficiency.

Building on these gaps and challenges, this study pursues three main objectives. The first is to determine whether ML models can accurately predict which teeth truly require extraction prior to RT, using only pre-RT clinical features. The focus is specifically on Logistic Regression (LR), Support Vector Machines (SVM), and Random Forests (RF). Second, to address the high dimensionality introduced by

OHE, this research develops two alternative approaches. This involves implementing categorical kernels to eliminate OHE entirely, with a specific focus on extending the kernel proposed by Belanche and Villegas (2013). In addition, it explores a less conventional technique by implementing a Categorical Autoencoder (CAE) to learn compact representations for classification. Third, this research investigates whether the performance of ensemble SVM strategies can be improved by adapting their standard implementation to utilize categorical kernels. Hence, the following research questions (RQ) will be explored:

**RQ1** Can ML models reliably predict tooth extraction necessity using limited pre-RT clinical data?

**RQ2** Which features are most important in the models for accurately classifying the need for tooth extraction?

**RQ3** Does the proposed extension of the categorical kernel by Belanche and Villegas (2013) improve performance over their original kernel?

**RQ4** Can a CAE provide a more effective alternative to OHE for training SVMs on high-dimensional categorical medical data?

**RQ5** Do ensemble SVMs with categorical kernels outperform ensembles using standard kernels or a single SVM?

This research offers contributions on both clinical and methodological fronts. From a clinical perspective, the developed ML models aim to support clinicians in making pre-RT extraction decisions, thereby reducing unnecessary procedures and improving patient QoL. Methodologically, the work advances the design of categorical kernels, evaluating them both as standalone alternatives to standard kernels and within ensemble SVM frameworks. In addition, by employing a CAE, the work further explores strategies for efficient representation learning in such settings. Together, these contributions provide a practical framework and potential guide for predictive modeling with high-cardinality categorical data.

# 2 Literature Review

This literature review is structured to address the knowledge gaps underlying this thesis's research questions. The review is divided into two main parts. First, on the clinical side, it establishes the context of ORN in HNC patients, examines existing studies on predicting dental extractions. Second, on the methodological side, it addresses the fundamental challenge of high-dimensionality when working with categorical features. This includes a review of research on categorical kernels for SVMs, an investigation of ensemble SVM strategies, and an exploration of alternative solution such as CAEs for dimensionality reduction.

## 2.1 Pre-Radiotherapy Tooth Extraction Decision-Making

ORN is a is a severe complication of RT. It is a condition in which the jawbone exposed to radiation fails to heal within 3 months which leads to leading to bone death (Chronopoulos et al. 2018). The risk of ORN starts at doses of 40 Gray (Gy) and increases with higher exposure, according to a study by Spijkervet et al. (2019). The Gy quantifies the absorbed radiation dose delivered to target tissues. To mitigate this risk, HNC patients to undergo dental examination before starting RT. This includes periodontal cleaning, extraction of teeth with poor prognosis, and treatment of oral infections in areas expected to receive radiation doses of 40 Gy or higher. At this stage, only diagnostic CT scans are available. These scans provide general information on the tumor and allow a rough estimation of the areas likely to be irradiated and the overall prescribed RT dose. Treatment planning CTs, which provide detailed dose distributions, are not yet performed. As a result, extraction decisions must be made under uncertainty.

Few studies have examined the necessity of pre-RT tooth extractions. Among them are Buurman et al. (2023) and Karaca et al. (2024), both note that a significant number of pre-RT extractions were unnecessary. For instance, Buurman et al. (2023) retrospectively reviewed HNC patients treated with RT between 2015 and 2019 at Maastro Clinic and Maastricht UMC+ (MUMC+). In their cohort, 1759 teeth were extracted prior to RT. Using the patients RT plans, the authors calculated the delivered mean dose ($D_{\text{mean}}$) and maximum point dose ($D_{\text{max}}$) for each extracted tooth to determine the actual radiation exposure. Applying a 40 Gy threshold, they found that 74% of extractions were unnecessary based on $D_{\text{mean}}$ and 61% based on $D_{\text{max}}$. Hence, studies such as Buurman et al. (2023) and Karaca et al. (2024) highlight the potential of ML models to predict tooth-specific radiation doses before extraction decisions. As such tools could significantly reduce unnecessary extractions and improve patient outcomes.

While predicting the need for dental extractions prior to RT is clinically significant, few studies have explored ML approaches to address this challenge. Chan et al. (2022) is one of the very few that tackles this issue. The study aimed to develop an AI model that predicts radiation doses to mandibular subsites before RT planning for oropharyngeal cancer, a subtype of HNC. Using 86 patient plans for training and 20 for testing, they implemented a regularized ensemble regression to classify whether subsites would exceed a $D_{\text{mean}}$ of 50 Gy, relying only on diagnostic CTs. On the test set, the model achieved a precision of 0.95 and a recall of 0.88, outperforming specialist oncologists. However, the models precision (0.82) was significantly higher than physicians' (0.25), while recall (0.94 vs. 1.00) was similar. However, the study is limited to predicting doses only for mandibular subsites, and its validation remains relatively small. Furthermore, it does not discuss which specific clinical features were most relevant for the predictions. These limitations directly motivate RQ1, which aims to build predictive models using broader clinical data, and RQ2, which specifically addresses feature importance for clinical understanding.

However, significant research has been conducted on AI and ML-based tooth extraction modeling in other patient groups, such as orthodontic planning and periodontal disease assessment. Most models typically analyze dental imaging along with clinical risk factors to predict whether a tooth requires extraction. Commonly used ML algorithms for this task, including LR, SVMs, and decision trees, have demonstrated accuracy rates ranging from 79% to 94% in these patient populations (Leavitt et al. 2023; Mason et al. 2023).

## 2.2 Limitations of One-Hot Encoding and Practical Alternatives

Building on the need for ML models to guide pre-RT tooth extractions, a key data pre-processing challenge must be addressed, namely, handling high-cardinality categorical variables. In many real-life applications, particularly in healthcare and insurance, datasets often contain high-cardinality categorical variables, features with hundreds or thousands of distinct values. Since most ML algorithms require numeric input, these variables must be encoded. While OHE is widely used due to its simplicity, it creates sparse, high-dimensional feature vectors for high-cardinality attributes. This sparsity increases computational cost and the risk of overfitting (Dahouda and Joe 2021; Liang 2025).

### 2.2.1 Dimensionality Reduction via Categorical Autoencoders

One strategy to overcome the dimensionality challenge of OHE is to compress the sparse matrix into a dense, lower-dimensional representation. Traditional methods like truncated Singular Value Decomposition, Principal Component Analysis project data linearly, but they often fail to capture non-linear relationships (Espadoto et al. 2021; Maćkiewicz and Ratajczak 1993). Neural network (NN) embeddings layers improve on this by learning dense vectors for each category during model training, thereby reducing memory usage and accelerating training compared to OHE (Guo and Berkhahn 2016). However, these embeddings are typically learned independently for each feature, ignoring interactions across different categorical variables. AEs provide an alternative for encoding high-dimensional variables, learning dense representations that capture non-linear feature interactions.

In a recent study, Delong and Kozak (2023) introduce a joint categorical AE architecture designed to encode all categorical features simultaneously through a single AE. The model takes the concatenated OHE inputs and, for each categorical feature, uses its own softmax output layer to predict a probability for every level. By jointly encoding categorical variables rather than processing them independently, their approach captures cross-feature dependencies. The authors evaluated their method on Poisson-distributed claim-frequency dataset, characterized by high-cardinality categorical risk factors along with continuous predictors. Their experiments demonstrate that using a single AE for all categorical features, rather than separate encoders, improved the NN's predictions, as the compressed representation from the AE's served as input to the final prediction model.

Building on the potential of categorical AEs, Carlin and Benjamini (2025) developed their own architecture to address a fundamental limitation of such models, namely, their dependence on OHE categorical features. This dependence becomes problematic with high-cardinality data, inflating dimensionality and slowing the training process. To overcome this, the authors introduce CardiCat, a variational AE (VAE) tailored for mixed-type tabular data with high-cardinality categorical columns. For categorical variables with three or more levels, CardiCat learns compact embeddings that are fed into the encoder and reconstructed by the decoder. Binary features remain in one-hot form, and numeric features pass through unchanged. The decoder architecture replaces large softmax output layers by predicting embeddings instead for high-cardinality features. This results in a reduction in parameters. The model was evaluated on seven datasets and demonstrated that CardiCat surpasses three competing VAE models in the reconstruction accuracy of the categorical features.

Together, these studies by Delong and Kozak (2023) and Carlin and Benjamini (2025) demonstrate that CAEs can effectively overcome the dimensionality challenges associated with OHE. However, their application remains primarily within NN frameworks. This motivates RQ4, which investigates whether the representations learned by a CAE can also enhance the performance and efficiency of SVMs on datasets with high-dimensional categorical features.

### 2.2.2 Categorical Kernels for Support Vector Machines

While the previous section focused on creating dense embeddings to be used in any model, an alternative strategy is to adapt the models themselves to handle categorical data natively. There has been significant progress in this direction for tree-based methods. However, few attempts have been made to extend this capability to other models, such as SVMs. For instance, relatively little research has focused on designing kernel functions that operate directly on categorical data.

SVMs were initially introduced for binary classification tasks, specifically for datasets consisting solely of continuous features. At its core, SVM aims to find a linear boundary in a kernel-transformed feature space, which relies on a distance measure. This creates a challenge when working with categorical features, as the distance between categorical values does not align with the distance measures used for continuous features. This problem motivated research that either designs kernels assigning sensible similarity scores to categorical values or completely avoid the explicit definition of a categorical distance altogether. However, despite its relevance, there is relatively little work on advancements in categorical kernels for SVMs.

Within the first line of work, two simple overlap-style kernels are commonly recommended, namely the Hamming kernel, which counts exact matches between categories, and the Jaccard kernel, which measures the proportion of shared categories relative to the union. Building on the Hamming kernel, Belanche and Villegas (2013) propose a probabilistic categorical kernel. Their key innovation is to weight matches by category rarity, so that matches on infrequent levels contribute more to the overall similarity. For categorical vectors $x_i$ and $x_j$, the kernel is defined as

$$k_1(x_i, x_j) = \exp\left(\frac{\gamma}{d} \sum_{k=1}^{d} k_1^{(U)}(x_{ik}, x_{jk})\right), \quad \gamma > 0$$

where $x_{ik}$ and $x_{jk}$ are the $k$th feature values for samples $i$ and $j$, $d$ is the number of categorical features and $\gamma$ controls the overall bandwidth of the kernel. It operates by applying a univariate kernel $k_1^{(U)}$ to each feature. This univariate kernel assigns a similarity of $(1 - P_Z(z)^\alpha)^{1/\alpha}$ for matches and zero for mismatches. The authors evaluated their probabilistic categorical kernel across three distinct datasets, comparing its performance against standard kernels. Belanche and Villegas (2013) reported it consistently outperformed the hamming, linear and rbf kernels, making it a strong alternative for categorical inputs. However, this kernel has the key limitation that different levels always receive zero similarity, so potential relationships between distinct categories are ignored.

The only study that pursued designing a kernel avoiding categorical distance altogether route is by Jung and Kim (2023). Their proposed kernel named SVM-C is tailored to work with many categorical or mixed features. The SVM-C dynamically reshapes the kernel space defined by the continuous features. The reshaping is guided by an effect measure, $\beta_j^g$, computed for each category $g$ of a feature $j$. This value quantifies whether a specific category is more associated with the positive or negative class, and by how much. During training, for every instance belonging to a category g, its position in the kernel space is adjusted. A positive $\beta_j^g$ shifts it to make a Class 1 prediction more likely, while a negative $\beta_j^g$ shifts it towards Class 0. Categorical features are processed sequentially based on their importance, $\kappa_j = \max_g |\beta_{jg}|$, which induces implicit variable selection, as the algorithm halts once further features no longer improve the model. SVM-C was evaluated against several SVM variants, standard SVMs with OHE as well as kernels that score categorical similarity. On 18 benchmark datasets, SVM-C generally matched or exceeded the performance of the other models, showing particular strength under class imbalance.

These studies demonstrate that categorical kernels can effectively handle categorical data, establishing them as a promising area for improving SVM performance. However, this review identified that the kernel proposed by Belanche and Villegas (2013) has a key limitation in its inability to model relationships between distinct categories. Hence, this provides a strong motivation for RQ3, which investigates whether a proposed extension improves upon the original Belanche and Villegas (2013) kernel.

## 2.3 Ensemble Support Vector Machine Architectures

The previous sections reviewed methods for handling categorical data within a single SVM model, through categorical kernels. A natural extension is to explore whether combining multiple such models can yield superior performance. Ensemble strategies, which combine multiple base learners, are widely used to improve the robustness, efficiency, and predictive performance of models, especially when dealing with a large number of variables. Common techniques include bagging, which generates diversity by training each base learner on a different bootstrap sample, and boosting, which sequentially constructs weak learners to correct the errors of previous models. These principles have been adapted to SVMs, resulting in architectures such as bagged SVMs and boosted SVMs.

Research on SVM ensembles has produced multiple studies. For instance, a comparative study on breast cancer prediction by Huang et al. (2017) evaluated single SVMs against ensembles using bagging and boosting with linear, polynomial, and rbf kernels. To evaluate the models, the study utilized two distinct datasets. Performance was assessed using accuracy, AUC/ROC, F-measure, and training time. The results indicated that while SVM ensembles generally provided slightly better predictive performance, the improvement was often insignificant. Similarly, S. Wang et al. (2009) conducted an extensive analysis across 20 datasets, evaluating bagged and boosted SVM ensembles of various sizes against single SVMs. A key finding was that bagging generally provided the best average accuracy, particularly when paired with polynomial kernels.

The ensemble concept has been further extended by methods that introduce randomness into the feature space, inspired by RFs. Building on this idea, Panov and Džeroski (2007) introduced the SubBag framework, which constructs ensembles by training each base learner on a bootstrap sample of the data combined with a random subset of features. As a general framework, SubBag can be applied to any base learner, enabling the creation of specific SVM variants. A possible variant is the Random Subspace SVM (RSM) which induces diversity by training each SVM on a randomly selected subset of the feature set. A direct implementation of the SubBag framework is the Random Subspace Bagged SVM (RSVM), which combines the data-level randomness of bagging with the feature-level randomness of RSM, training each model on both a bootstrap sample and a random feature subspace. A study by Bos (2024) directly evaluated this SubBag approach for SVMs. To evaluate the RSVM model, Bos (2024) conducted experiments on five benchmark datasets using three kernel functions, namely, the linear, polynomial, and rbf kernels. For each dataset–kernel combination, a single SVM was trained on the full feature set, additionally, bagged, RSM and RSVM ensembles were constructed. These ensembles varied over a range of ensemble sizes $B$ and feature-subset sizes $m$. Across all configurations, RSVM and RSM yielded almost identical accuracy values, with neither method consistently outperforming the other. Both ensemble approaches generally matched or slightly outperformed the performance of bagged SVMs. Furthermore, in most cases, they also exceeded the performance of standalone SVMs, although in certain settings a single SVM trained on the full feature set achieved the highest accuracy.

This section reveals that while studies have extensively evaluated SVM ensembles with standard kernels, no research has been conducted on the performance of ensemble architectures when utilizing kernels specifically designed for categorical data. This gap directly motivates RQ5, which investigates whether integrating categorical kernels into these ensemble frameworks can yield superior performance.

# 3 Methodology

This section outlines the principles of the ML algorithms used in this study. It then describes the hyperparameter tuning strategy and the corresponding search spaces, followed by the evaluation metrics for model assessment. The section concludes with an overview of the interpretability techniques employed.

## *3.1 Logistic Regression*

LR is a widely-used supervised ML algorithm used for binary classification. Given an input vector $X = (x_1, x_2, \ldots, x_p)$, the model estimates the probability $p(X) = \Pr\big(Y = 1 \mid X\big)$ that the "positive" class $Y = 1$ occurs.

LR assumes that the log-odds of the event $Y = 1$ is a linear combination of the input features $X$ (Starbuck 2023). Concretely, this can be written as:

$$\log \frac{p(X)}{1 - p(X)} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

where $\beta = (\beta_0, \beta_1, \ldots, \beta_p)$ are parameters to be estimated.

To convert the log-odds into a probability in the range $[0, 1]$, the sigmoid function is applied. Thus the model's predicted probability is expressed as follows:

$$p(X) = P(Y = 1 \mid X) = \frac{\exp(\beta_0 + \beta^T X)}{1 + \exp(\beta_0 + \beta^T X)} = \sigma\big(\beta_0 + \beta^T X\big)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function.

Finally, a threshold $t$, which is commonly set to 0.5, is applied to $p(X)$ to decide the class label:

$$\hat{Y} = \begin{cases} 1, & p(X) \geq t, \\ 0, & p(X) < t. \end{cases}$$

where $\hat{Y} = 1$ indicates that the event that $Y = 1$ occurs, and $\hat{Y} = 0$ indicates that it does not.

During training, LR estimates the parameters $\beta$ such that the predicted probabilities align as closely as possible with the observed outcomes. This is achieved by maximizing the likelihood of the training labels given the inputs, or equivalently by minimizing a cost function derived from this likelihood (Joshi and Dhakal 2021). The corresponding log-likelihood is expressed as:

$$\mathcal{L}(\beta) = \sum_{i=1}^{n} [y_i \log p(x_i) + (1 - y_i) \log(1 - p(x_i))]$$

where $y_i \in \{0, 1\}$ are the true labels.

## 3.2 Random Forest

RF is an ensemble method in the supervised learning family used for both classification and regression tasks. It works by building multiple classification and regression trees (CARTs) and combining their output to make predictions (Fawagreh, Gaber, and Elyan 2014). For classification tasks, the final decision is made by majority voting across the trees. RF combines two fundamental mechanisms, bagging and random feature selection, which significantly reduces the variance that comes from each individual CART model.

Bagging refers to the process of creating $B$ bootstrapped samples of size $N$, sampled with replacement, from the original training set (Z. Wang et al. 2018). Subsequently, a decision tree is trained on each of these $B$ samples. The instances that were omitted from the bootstrapped samples are referred to the out-of-bag instances, which provide an unbiased estimate of the generalization error. Additionally, at each split within a tree, only a randomly chosen subset of features is considered, which further enhances the generalization of the model.

Since each tree in the forest is a CART model, splits are chosen to maximize node purity according to the Gini impurity criterion (Sun et al. 2024). The Gini impurity for a node $n$ with $J$ classes is given by

$$\text{Gini}(n) = 1 - \sum_{j=1}^{J} p_j^2$$

where $p_j$ is the proportion of class $j$ in the node $n$. At each candidate split, the feature that produces the largest drop in Gini is selected (Sarica, Cerasa, and Quattrone 2017).

Mathematically the RF model can be expressed as $R = \{h(x, \beta_k), k = 1, 2, \ldots, k\}$. Here, each $h(x, \beta_k)$ represents a base classifier, x represents the original data set, and each $\beta_k$ is the $k$-th bootstrap sample drawn from x via the bagging procedure. Lastly, $k$ denotes the number of trees in the RF. To output class probabilities, the RF aggregates the probabilities predicted by all trees in the ensemble (Scikit-learn 2014). For a given tree $k$, the probability is estimated as the proportion of samples belonging to each class in the leaf node where $x_i$ falls. The final probability is then the average across all $B$ trees.

## 3.3 Support Vector Machines

SVMs are supervised learning algorithms used for both classification as well as regression. In the classification setting, a SVM attempts to find a hyperplane that separates the classes as well as possible. This is achieved by maximizing the margin, which is the distance between the hyperplane and the nearest data points from each class, known as support vectors (Bhavsar and Panchal 2012).

The hyperplane can be mathematically expressed as

$$H(x) = w^{\top}x + b = 0$$

where $x$ is the input feature vector, $w$ refers to a weight vector and $b$ the bias term which are all parameters to be learned.

The margin is defined as the sum of the distances from the hyperplane to the closest positive and negative samples, denoted $d_+$ and $d_-$, respectively (Zayrit et al. 2021). For an ideal linear SVM, these distances are equal, which yields

$$\text{Margin} = d_+ + d_- = \frac{2}{\|w\|}.$$

To ensure that every pair $(x_i, y_i)$ is correctly classified and lies outside the margin, the SVM imposes the constraint:

$$y_i(w^{\top}x_i + b) \geq 1 \quad \forall i,$$

where $y_i \in \{-1, +1\}$ indicates the class label. This formulation flips the inequality depending on the class, allowing it to be written compactly as a single condition. Hence, finding the optimal separating hyperplane can be expressed as the following optimization problem:

$$\begin{aligned} &\min_{w,b} \quad \frac{1}{2}\|w\|^2 \\ &\text{subject to} \quad y_i(w^{\top}x_i + b) \geq 1 \quad \forall i \end{aligned}$$

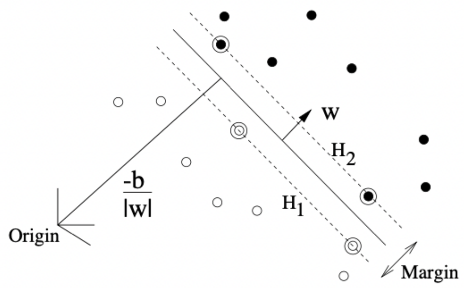



Figure 1: Representation of the SVM's kernel trick (Pimentel, Ospina, and Ara 2021)

However, this formulation assumes that the data is linearly separable and every data point falls on the on the correct side of the hyperplane. In practice, this is often not the case due to outliers or overlapping classes. To generalize SVMs to handle such cases, two key extensions are introduced, namely, slack variables and the kernel trick.

In the presence of outliers or non-seperable data, SVM can work around this by introducing positive slack variable ($\zeta_i, i = 1, \ldots, m$) which allows for some misclassification. The problem is then rewritten as:

$$
\begin{aligned}
&\min_{w,b,\{\zeta_i\}} \quad \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{m}\zeta_i \\
&\text{s.t.} \qquad y_i\big(w^T x_i + b\big) \geq 1 - \zeta_i, \quad i = 1, 2, \ldots, m, \\
&\qquad\qquad \zeta_i \geq 0, \quad i = 1, 2, \ldots, m
\end{aligned}
$$

Here, $C > 0$ is the penalty weight on slack variables, with larger $C$ imposing heavier penalties on misclassifications.

While slack variables allow the SVM to handle imperfectly separable data, they do not address scenarios where the data is not linearly separable in the input space at all. To address this, the SVM optimization problem can be rewritten in its Lagrangian form using multipliers $\alpha_i$:

$$L_P \equiv \frac{1}{2}\|w\|^2 - \sum_{i=1}^{l} \alpha_i y_i \big(x_i \cdot w + b\big) + \sum_{i=1}^{l} \alpha_i$$

This Lagrangian reformulation replaces the original inequality constraints, coupling $w$ and $b$ to every training point $x_i$ with the following simple conditions on the multipliers

$$0 \le \alpha_i \le C, \qquad \sum_{i=1}^{l} \alpha_i y_i = 0.$$

Substituting these relations into the dual yields a problem that depends on the data only through inner products $x_i^\top x_j$. This enables nonlinear decision boundaries by replacing each inner product with a kernel

$$K(x_i, x_j) = \phi(x_i)^\top \phi(x_j),$$

where $\phi : \mathbb{R}^d \to \mathcal{H}$ maps inputs into a possibly infinite-dimensional feature space $\mathcal{H}$. The kernelized dual is

$$\begin{aligned} \text{maximize} \quad & L_D(\alpha) = \sum_{i=1}^{l} \alpha_i - \frac{1}{2} \sum_{i=1}^{l} \sum_{j=1}^{l} \alpha_i \alpha_j y_i y_j K(x_i, x_j) \\ \text{subject to} \quad & 0 \le \alpha_i \le C \quad \text{for } i = 1, \ldots, l, \\ & \sum_{i=1}^{l} \alpha_i y_i = 0. \end{aligned}$$

The resulting decision function is

$$f(x) = \sum_{i=1}^{l} \alpha_i^* y_i K(x_i, x) + b,$$

which corresponds to a linear separator in $\mathcal{H}$ and a nonlinear separator in the input space.

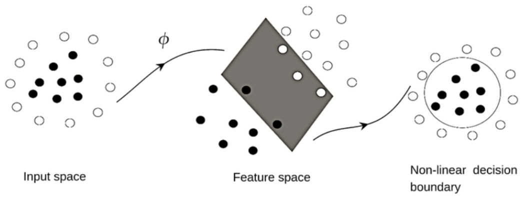



Figure 2: Linear separating hyperplanes for a separable dataset, with support vectors circled (Burges 1998)

Common choices of $K$ include the polynomial kernel $K(x,z) = (x^\top z + c)^d$, the rbf kernel $K(x,z) = \exp(-\gamma\|x - z\|^2)$, and the sigmoid kernel $K(x,z) = \tanh(\kappa x^\top z + \theta)$.

Rather than only outputting a binary label $y \in \{0,1\}$, the SVM's predictions can also be converted into probabilities $P(Y = 1 \mid x_i)$ via Platt scaling. This involves fitting two parameters, $A$ and $B$, so that

$$P\big(Y = 1 \mid x_i\big) = \frac{1}{1 + \exp\big(A f(x_i) + B\big)},$$

where

$$f(x_i) = w^\top x_i + b$$

denotes the SVM decision function. The resulting sigmoid transformation maps those scores to class probabilities rather than hard labels.

### 3.3.1 Categorical Kernel

In addition to standard kernels for continuous variables, this study moves employs categorical kernels to eliminate the need for OHE. Specifically, the categorical kernel $k_1$ proposed by Belanche and Villegas (2013) is employed and extended. The main limitation of $k_1$ kernel is the "zero-match" problem, where any two distinct categories are assigned zero similarity. The extension, named $k_2$, overcomes this by assigning non-zero, rarity-aware similarity to distinct categories and incorporates a numerical component to handle mixed data types.

The $k_1$ kernel is a probabilistic kernel designed to improve upon the classical Hamming similarity. The Hamming kernel, $k_0$, compares two categorical variables and returns 1 if they are equal and 0 otherwise. For $d$ categorical variables, this kernel is defined as:

$$k_0(x_i, x_j) = \frac{1}{d} \sum_{k=1}^{d} \mathbb{I}_{\{x_{ik}=x_{jk}\}}$$

where $\mathbb{I}_{\{x_{ik}=x_{jk}\}}$ denotes the indicator function, which equals 1 if $x_{ik} = x_{jk}$ and 0 otherwise. Here, $x_{ik}$ is the value of the $k$-th categorical feature of sample $i$ and $x_{jk}$ is the value of the $k$-th categorical feature of sample $j$. While mathematically valid, this kernel treats all matches equally and does not account for how rare or common a category is.

The $k_1$ kernel addresses this by reweighting matches based on their empirical frequency. It is constructed in two stages. First, a univariate kernel is defined for a single categorical feature:

$$k_1^{(U)}(z_i, z_j) = \begin{cases} h_\alpha(P_Z(z_i)) & \text{if } z_i = z_j \\ 0 & \text{if } z_i \neq z_j \end{cases}$$

Here, $P_Z(z)$ denotes the probability of observing category $z$, and $h_\alpha$ is a weighting function that emphasizes rarer matches. It is defined as:

$$h_\alpha(P_Z(z)) = \left(1 - P_Z(z)^\alpha\right)^{1/\alpha}, \quad \alpha > 0$$

This transformation downweights frequent categories while emphasizing rare ones, so matches on low-probability values yield higher kernel similarity. As $\alpha$ approaches 0, the function converges to a constant, making the kernel less discriminative. This approach provides a more expressive similarity measure for categorical data that captures underlying distributional information. Second, the univariate kernels are combined over all $d$ categorical features into a multivariate kernel using an exponential formulation:

$$k_1(x_i, x_j) = \exp\left(\frac{\gamma}{d} \sum_{k=1}^{d} k_1^{(U)}(x_{ik}, x_{jk})\right), \quad \gamma > 0$$

where $x_{ik}$ and $x_{jk}$ are the $k$th feature values for samples $i$ and $j$, and $\gamma$ controls the overall bandwidth of the kernel.

To overcome the zero-match limitation, this study generalizes the univariate kernel to:

$$k_n^{(U)}(z_i, z_j) = \begin{cases} h(P_Z(z_i)) & \text{if } z_i = z_j \\ g_n P_Z(z_i), P_Z(z_j)) & \text{if } z_i \neq z_j \end{cases}$$

where the function $h_\alpha$ is as previously defined. The critical innovation is the introduction of the off-diagonal term $g(P_i, P_j)$. The $k_1$ kernel uses $g(P_i, P_j) = 0$ for distinct categories. Whereas the function $g(P_i, P_j)$ for $k_2$ is defined as:

$$g_2(P_i, P_j) = \exp\left(-\frac{(P_i - P_j)^2}{\sigma^2}\right) \sqrt{h_\alpha(P_i)\, h_\alpha(P_j)}$$

In $k_2$, the parameter $\sigma > 0$ controls how quickly the off-diagonal similarity decays with increasing frequency difference. When $\sigma$ approaches 0, the exponential term $\exp\left(-\frac{(P_i - P_j)^2}{\sigma^2}\right)$ vanishes for $P_i \neq P_j$, and $k_2$ reduces to $k_1$. This formulation ensures that pairs of rare categories receive relatively high similarity, while common–common pairs receive low similarity.

For categorical vectors, ($x_i$ and $x_j$), the multivariate kernel is defined as:

$$k_n(x_i, x_j) = \exp\left(\frac{\gamma}{d} \sum_{k=1}^{d} k_n^{(U)}(x_{ik}, x_{jk})\right), \quad \gamma > 0.$$

where $n \in \{1, 2\}$ selects the univariate kernel $K_n^{(U)}$ for the individual categorical features.

In the presence of both categorical and continuous input features, the overall kernel function is extended by combining the categorical kernel with a standard rbf kernel applied to the continuous features. Specifically, the final kernel is defined as:

$$k_n(x_i, x_j) = k_{\text{cont}}(x_i^{(\text{cont})}, x_j^{(\text{cont})}) + k_n(x_i^{(\text{cat})}, x_j^{(\text{cat})})$$

where $x_i^{(\text{cont})}$ and $x_j^{(\text{cont})}$ denote the continuous components of the samples $x_i$ and $x_j$, respectively, and $k_{\text{cont}}$ is the standard rbf kernel:

$$k_{\text{cont}}(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right)$$

This additive formulation allows the kernel to jointly capture similarity across both categorical and continuous variables in a unified manner.

### 3.3.2 Random Subspace Support Vector Machine

In addition to single SVM models, this study implements ensemble methods to enhance performance and robustness. The first ensemble approach is the RSM. This model is an ensemble method based on the Random Subspace Method proposed by Ho (1998). This method is applicable to arbitrary base learners, where each model is trained on a randomly selected subset of features.

Specifically, given a training set with $d$ features, it builds each base model on a full set of $N$ training examples but projects those examples onto a randomly chosen subset of features of size $m \leq d$. Panov and Džeroski (2007) noted that this strategy is especially advantageous when the number of features greatly exceeds the number of observations or when many predictors are redundant, since it reduces the dimensionality without throwing away any instances. In the SVM setting, each member of the ensemble is trained with the kernel $k$ on the selected subset of features. Its hyperparameters are tuned via grid search before being added to the ensemble. The exact RSM algorithm, as presented by Bos (2024), is reproduced in Appendix C.

### 3.3.3 Random Support Vector Machine

The RSVM implements the SubBag framework introduced by Panov and Džeroski (2007), adapting it specifically for SVMs. The key idea of this method is to generate an ensemble $E$ of $B$ models, where each model is trained on both a bootstrapped sample of the data and a random selection of the features.

Following the implementation of Bos (2024), at each iteration $b$, a base SVM with kernel $k$ is trained on the corresponding data subset. The hyperparameters for each SVM in the ensemble are individually tuned via grid search before being added to $E$. The complete algorithm is reproduced in Appendix D.

## 3.4 Categorical Autoencoder for Dimensionality Reduction

The AE in this study serves primarily as a preprocessing tool, transforming categorical features into a compact representations. This compressed representation can then be used as input for any model, such as the SVMs employed in this work. An AE is a special type of NN that is designed to compress input data into a compact, informative representation and then reconstructs the original input from this compressed form, aiming to make the output as close to the input as possible (Bank, Koenigstein, and Giryes 2023). It consists of two main components, namely, the encoder and decoder. The encoder is in charge of extracting the important features and mapping the input data into a reduced dimensionality. Whereas the decoder rebuilds the data from this compressed representation.

Its architecture consists of three layers (see Figure 3), the input layer, the hidden layer and output layer. First, the input layer, which contains $n$ neurons, corresponding to the dimensionality of the input data, accepts the raw input data. Next comes the hidden layer, often called the bottleneck, this layer comprises $t$ neurons. Whenever $t < n$, it enforces the network to learn a compressed representation that captures only the most important features. Finally, the output layer, also with $n$ neurons, is responsible for reconstructing the original input.

As seen on Figure 3 the input layer together with the hidden layer form the encoder, whereas the hidden layer and the output layer form the decoder. In the encoding stage, the AE receives the original data $X$, and encoding produces the lower dimensional data $h$ through a mapping:

$$h = f(W_1 X + b_1),$$

where $W_1$ represents the weight matrix between the input layer and the hidden layer and $b_1$ is the bias vector. Lastly, $f$ denotes the activation function, which refers to the transformation applied to the weighted inputs that allows the network

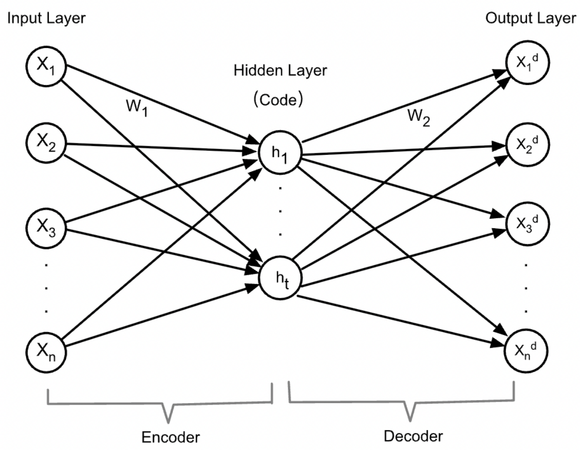



Figure 3: The structure diagram of an AE (P. Li, Pei, and J. Li 2023)

to capture complex patterns. Once $h$ has been computed, the decoding stage reconstructs the original data by applying the following transformation:

$$X^{dec} = g(W_2 h + b_2),$$

here, $W_2$ represents the weight matrix between the hidden layer and the output layer, $b_2$ is the bias vector and $g$ denotes the activation function.

The parameters $\theta = \{W_1, b_1, W_2, b_2\}$ of the AE are learned by minimizing a reconstruction loss $J_{\text{AE}}(\theta)$ between the decoded output $X^{dec} = \{x_1^{dec}, x_2^{dec}, \ldots, x_n^{dec}\}$ and the original input $X = \{x_1, x_2, \ldots, x_n\}$. When inputs are of continuous nature, the mean squared error loss is used, hence the objective is

$$J_{\text{AE}}(\theta) = \frac{1}{2} \sum_{i=1}^{n} \|x_i^{dec} - x_i\|^2.$$

where $x_i$ denotes the $i$-th input example. For binary inputs, the binary cross-entropy loss is utilized

$$J_{\text{AE}}(\theta) = -\sum_{i=1}^{n} \big[x_i \log(x_i^{dec}) + (1 - x_i) \log(1 - x_i^{dec})\big],$$

Minimizing $J_{\text{AE}}(\theta)$ trains the encoder–decoder pair to reconstruct the data under the chosen metric. Furthermore, the optimization is performed via stochastic gradient descent, which iteratively updates the parameters $\theta$ in the direction of steepest descent to reduce the reconstruction error.

To construct the CAE, the architecture from Delong and Kozak (2023) is adopted. This AE is specifically designed to learn a compact, low-dimensional representation of OHE categorical variables and to reconstruct the original OHE vectors from this representation.

Given $d$ categorical variables $x = \left(x^{(1)}, \ldots, x^{(d)}\right)$, where each $x^{(j)}$ has $m_j$ distinct labels, the first step is to transform $x$ into its OHE form. Hence, variable $x^{(j)}$, which takes one of those $m_j$ labels, is represented by an $m_j$-dimensional binary vector that contains a single 1 at the position of the observed category and 0s elsewhere.

Concatenating these one-hot vectors yields the full binary input $x^{\text{cat}}$ to the AE:

$$x^{\text{cat}} = \left((x_1^{\text{cat}})^\top,\ (x_2^{\text{cat}})^\top,\ \ldots,\ (x_d^{\text{cat}})^\top\right)^\top \in \{0,1\}^M, \quad M = \sum_{j=1}^{d} m_j.$$

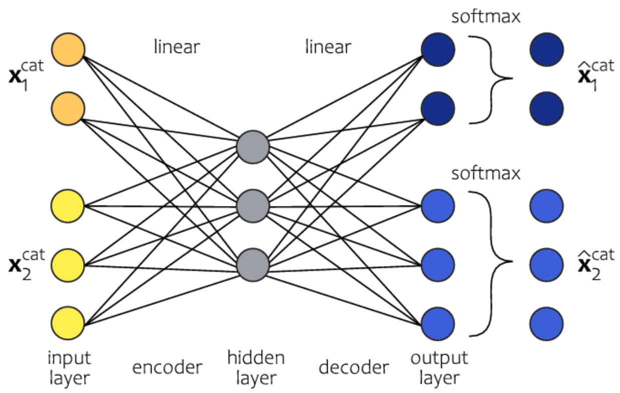



Figure 4: Architecture of the AE of type Joint AE for categorical features (Delong and Kozak 2023)

The joint AE network of Delong and Kozak (2023) consists of three successive layers (see Figure 4). The input layer of size $M$ applies a linear activation without bias, passing $x^{\text{cat}}$ directly through. The second layer is a bias-free linear transformation of size $l$, yielding a compressed representation

$$h = W_1 x^{\text{cat}} \in \mathbb{R}^l, \quad b_1 = 0$$

A second linear layer in the decoder transforms the hidden embedding $h \in \mathbb{R}^l$

into $M$ raw scores, or called logits:

$$z = W_2 h + b_2 = (z_1, \ldots, z_M) \in \mathbb{R}^l.$$

Here, $x^{\text{cat}} \in \mathbb{R}^M$ denotes the OHE categorical input, $W_1 \in \mathbb{R}^{l \times M}$ and $W_2 \in \mathbb{R}^{M \times l}$ are the encoder and decoder weight matrices, respectively. Furthermore $b_1$ and $b_2$ are bias terms, with $b_1 = 0$ by design.

These $M$ logits are partitioned into blocks corresponding to each original categorical feature. For instance, the first $m_1$ entries $(z_1, \ldots, z_{m_1})$ for feature 1, the next $m_2$ entries $(z_{m_1+1}, \ldots, z_{m_1+m_2})$ correspond to feature 2, and so on. Within each block, the softmax function is applied independently. For feature $j$, whose block spans indices $r = \tilde{m}_{j-1} + 1, \ldots, \tilde{m}_j$ (with $\tilde{m}_j = \sum_{u=1}^{j} m_u$), the probabilities are

$$\pi_r(x^{\text{cat}}) = \frac{\exp(z_r)}{\sum_{s=\tilde{m}_{j-1}+1}^{\tilde{m}_j} \exp(z_s)} \quad (r = \tilde{m}_{j-1} + 1, \ldots, \tilde{m}_j).$$

Concatenating all per-feature probability vectors $\left(\pi_1, \ldots, \pi_M\right)$ yields the reconstructed categorical output $\hat{x}^{\text{cat}} = \pi(x^{\text{cat}}) \in \mathbb{R}^M$.

Lastly, the network is trained by minimizing the sum of cross-entropy losses over all categorical features. For each example $i = 1, \ldots, n$, let $x_i^{\text{cat}} \in \{0, 1\}^M$ be its one-hot input and $\pi(x_i^{\text{cat}}) \in [0, 1]^M$ the reconstructed probabilities. The loss is then

$$L\big(\pi(x_i^{\text{cat}}), x_i^{\text{cat}}\big) = -\sum_{j=1}^{c} \sum_{r=1}^{m_j} x_{j_r,i}^{\text{cat}} \log\big(\pi_r(x_i^{\text{cat}})\big), \quad i = 1, \cdots, n.$$

This criterion encourages each per-feature softmax to assign probability one to the true category.

## 3.5 Hyperparameter Tuning

In this study, the method chosen to identify the optimal hyperparameter settings is GridSearch. This approach systematically tries every combination of the specified candidate parameters during training, evaluating each one using cross-validation (CV) to ensure robust performance estimates. It then selects the combination that yields the best average results across the folds. The specific hyperparameters and their tested values depend on the particular ML algorithm. For each model, the CV setting and a summary of all hyperparameters are displayed in the Appendix E. In general, a 5-fold CV was used for the baseline models (LR, SVM, RF). For the ensemble SVMs a 3-fold split is used to manage computational cost.

Furthermore, to address class imbalance and prioritize correct identification of extractions, several class-weight schemes (balanced, {0:1,1:2}, {0:1,1:4}) are also implemented during GridSearch. The balanced weighting scheme equalizes the contribution of each class to the loss, while custom schemes double or quadruple the penalty for misclassifying an extraction.

## 3.6 Calibration Methods

Although ML methods like RF or SVMs can output class probabilities, these raw estimates often fail to reflect true event frequencies. However, this problem can be resolved through model calibration. This is a process that adjusts a model's scores so that its predicted probabilities more closely match observed outcome rates (Scikit-Learn 2024). The approach used in this study is Platt scaling which fits a sigmoid function to the classifier's decision scores:

$$P_{\text{cal}}(y = 1 \mid x) = \frac{1}{1 + \exp\big(A f(x) + B\big)},$$

where $f(x)$ is ML models prediction, and $A$ and $B$ are scalar parameters learned by maximizing the likelihood on held-out data. This is implemented using Scikit-Learn, the CalibratedClassifierCV wrapper automates this process by performing $k$-fold CV on the training set. It fits the base estimator on each fold's training split, then estimates $A$ and $B$ on the corresponding validation split. At prediction time, the calibrated probabilities from each fold are averaged, yielding final outputs that are far better aligned with true outcome frequencies.

## 3.7 Evaluation Metrics

The model evaluation is based on categorical forecasts or probabilistic forecasts, which lead to two types of performance metrics, namely, hard scoring functions and probabilistic scoring functions.

Hard metrics are used when the model outputs only a class label. These are based on the four fundamental outcomes of a binary classifier: True Positive (TP), when the model correctly labels a positive sample; True Negative (TN), when a negative sample is correctly identified False Positive (FP), when a negative sample is wrongly labeled as positive; and False Negative (FN), when a positive sample is wrongly labeled as negative. These outcomes form the confusion matrix shown in Table 1.

Table 1: Confusion Matrix for Binary Classification

| | **Actual Positive Class** | **Actual Negative Class** |
|---|---|---|
| **Predicted Positive Class** | TP | FN |
| **Predicted Negative Class** | FP | TN |

From this matrix, standard evaluation metrics can be derived. This study employs the following key metrics:

- Precision ($P$): measures the proportion of predicted positives that are actually correct, given by
$$P = \frac{TP}{TP + FP}$$
- Recall ($R$): measures the proportion of actual positives that the model correctly identifies, given by
$$R = \frac{TP}{TP + FN}$$
- F1-Score ($F_1$): the harmonic mean of precision and recall, balancing both metrics, given by
$$F_1 = \frac{2\,(P \times R)}{P + R}$$

- Accuracy (Acc): the fraction of all predictions that are correct, given by

$$\mathrm{Acc} = \frac{TP + TN}{TP + TN + FP + FN}$$

In this clinical context the cost of missing a tooth that truly requires extraction, in other words a FN, is far greater than that of performing an unnecessary extraction. Therefore, the model's performance is prioritized based on its recall, as it directly measures the model's ability to avoid missing necessary extractions.

When the models output predicted probabilities instead of hard labels, probabilistic scores are used. The most commonly used metrics in this scenario are the Brier Score (BS) and the logarithmic loss. The BS is defined as

$$\mathrm{BS} \; = \; \frac{1}{n} \sum_{i=1}^{n} (p_i - y_i)^2$$

where $p_i \in [0, 1]$ is the predicted probability $p_i = P(y_i = 1 \mid x_i)$ for instance $i$ and $y_i \in \{0, 1\}$ is the corresponding true outcome. A lower BS indicates more accurate and better-calibrated probabilistic forecasts.

The logarithmic loss also known as cross-entropy, quantifies the discrepancy between true binary labels $y_i \in \{0, 1\}$ and predicted probabilities. However, standard logarithmic loss can be misleading on imbalanced datasets as the majority class will dominates the score, masking performance on the minority class. To address this, one can use the balanced cross–entropy (bCE) metric, which introduces a class–weighting factor $\alpha \in [0, 1]$.

$$\mathrm{bCE} = -\frac{1}{n} \sum_{i=1}^{n} \Big[ \alpha \, y_i \, \log_2(p_i) \; + \; (1 - \alpha)(1 - y_i) \, \log_2(1 - p_i) \Big]$$

# 4 Data Overview and Preparation

The primary dataset for this study comprises data from 399 HNC patients treated with RT at the Maastro Clinic and Maastricht University Medical Center+ (MUMC+) between 2015 and 2019. In total, 1789 teeth were extracted prior to RT. While all extracted teeth were clinically compromised (e.g., by decay, periodontal disease, or structural damage), not all would necessarily have required removal when assessed against radiation-based risk thresholds.

Patients were excluded from the dataset if they were edentulous, had no pre-RT extractions, underwent extractions only after RT, required full-mouth extractions due to severe dental neglect, or had a history of prior HNC RT, proton therapy, brachytherapy, or palliative RT. Additional exclusions were made for tumors of unknown primary origin or bilateral involvement. The dataset is organized at the patient level, where each row corresponds to a single patient and includes demographic, tumor, treatment, and tooth-related variables collected during dental screening. A summary is provided in Table 2.

Table 2: Summary of variables available at dental screening

| **Variable Group** | **Variables** |
|---|---|
| Demographic | Age, Sex |
| Tumor-related | TumorLaterality, TumorLocation, TumorCategory, TNM_T, TNM_N, TNM_Stage |
| Treatment-related | TotalDose, SystemicTherapy |
| Tooth-related | ToothType, nExtracted, ExtractionFlag_dmean, ExtractionFlag_dmax |

The demographic group contains general patient information, specifically age and sex. The tumor-related group includes variables that describe the tumor's characteristics. TumorLaterality indicates which of the four quadrants (left/right, upper/lower) the tumor occupies, whereas TumorCategory specifies its type. TNM_T describes the the size of the main tumor, while TNM_N refers to the spread to nearby lymph nodes. The variable TNM_Stage combines T and N into a single overall stage, it also specifies whether there is distant metastasis. TumorLocation and TumorSite specify the anatomical region affected. Lastly, Stage and StageItmIV provide additional staging detail which help distinguish advanced disease. Finally, the tooth-related group includes the variables ToothType, indicating which tooth is under consideration, and nExtracted, denoting the number of teeth extracted prior to RT. It also includes two binary flags, ExtractionFlag_dmean and ExtractionFlag_dmax, that indicate whether extraction was necessary based on the mean $D_{\text{mean}}$ or maximum $D_{\text{max}}$ radiation dose.

## 4.1 Descriptive Analysis of the Tooth Extraction Dataset

First, the frequencies of the categorical variables, excluding ToothType, are summarized in Table 3. Since the number of levels for ToothType was high, it was more efficient to visualize its distribution (see Figure 5) rather than present it in the frequency table. In terms of demographics, the dataset comprised 399 patients, of whom the majority were male (70%).

With respect to tumor characteristics, TumorLaterality was nearly evenly split between Laterality 1 (46%) and Laterality 2 (47%), wheras Laterality 3 (3%) and 4 (3%) were relatively uncommon. For TumorLocation, the most frequent sites were Location 3 (31%) and Location 4 (28%), with the remaining locations each representing smaller proportions (<14%). Furthermore, TumorSite showed a similar distribution as the TumorLocation feature, with Site 3 (30%) being the most common, followed by Site 5 (17%) and Site 8 (16%), while all other sites occurred in fewer than 13% of patients.

The feature TNM_T showed that T3 is the most common stage (29%), followed by T2 (26%) and T4 (22%). In terms of nodal staging (TNM_N), 40% of patients were classified as N0, with decreasing frequencies for N1 (22%), N2 (25%), and N3 (13%). Overall TNM Staging indicated that Stage IV was the most prevalent (40%), while earlier stages (I-III) were less common, ranging between 17% and 26%. A comparable pattern was observed for the ItmIV staging variable, where Stage IV again dominated (40%), while very few patients were classified in Stage 0 (0.3%), and Stages I–III were more evenly distributed (17–25%). Furthermore, the variable Stage showed that Stage 4 was the most frequent (26.3%), followed by Stage 3 (24.6%) and Stage 2 (16.5%), whereas advanced categories such as Stage 5 (13.0%) and rare cases of Stage 6 (0.3%) were less common. Lastly, for the variable TumorCategories, the majority of cases (81.5%) were classified as Type 1, while all remaining categories were comparatively rare, each representing less than 10% of the sample.

Furthermore, most patients received Therapy 4 (64%) as systemic treatment. Therapy 0 was the next most frequent (27%), while all other therapies (Types 1, 2, 5, and 6) were in fewer than 8% of patients combined. Lastly, missing data is minimal, generally under 3% across all variables. Overall, the frequency analysis reveals clear imbalances across categorical variables, with certain levels strongly dominating the distributions.

Lastly, the distribution of extracted teeth was analyzed to identify which tooth types were removed most frequently. Teeth are numbered according to the World Dental Federation notation. A full overview of the numbering scheme is provided in Appendix B. As shown in Figure 5, molars, particularly teeth 37, 36, and 47, were extracted most often, while anterior teeth such as 14, 18, and 21 were among the least frequently extracted.

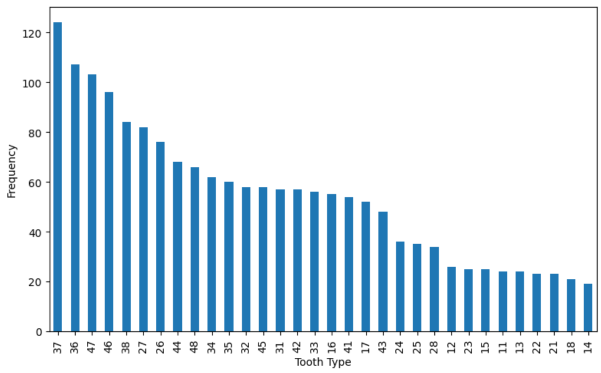



Figure 5: Frequency distribution by ToothType

Next, descriptive statistics for all numerical variables in the dataset are summarized in Table 4. In addition, four new features were derived, specifically, nJustified_mean, nUnjustified_mean, nJustified_max, and nUnjustified_max, representing, for each patient, the number of necessary and unnecessary extractions based on $D_{\mathrm{mean}}$ and $D_{\mathrm{max}}$. These features were obtained by comparing the radiation dose received by each tooth, according to either $D_{\mathrm{mean}}$ or $D_{\mathrm{max}}$, with the 40 Gy threshold, and then summing the results across teeth. In addition, the table indicates that patients ranged in age from 22 to 93 years, with the central 50% of values

Table 3: Frequencies of Categorical Variables ($n = 399$) in the Tooth Extraction Dataset

| **Variable** | **n (%)** |
|---|---|
| *Sex* | |
| Male | 279 (69.92) |
| Female | 120 (30.08) |
| *Tumor Laterality* | |
| Laterality 1 | 185 (46.37) |
| Laterality 2 | 186 (46.62) |
| Laterality 3 | 13 (3.26) |
| Laterality 4 | 15 (3.76) |
| *Tumor Location* | |
| Location 1 | 54 (13.53) |
| Location 2 | 16 (4.01) |
| Location 3 | 122 (30.58) |
| Location 4 | 110 (27.57) |
| Location 5 | 26 (6.52) |
| Location 6 | 11 (2.76) |
| Location 7 | 35 (8.77) |
| Location 8 | 25 (6.27) |
| *Tumor Stage (TNM_T)* | |
| T1 | 65 (16.29) |
| T2 | 105 (26.32) |
| T3 | 114 (28.57) |
| T4 | 88 (22.06) |
| T5 | 27 (6.77) |
| *Node Stage (TNM_N)* | |
| N0 | 160 (40.10) |
| N1 | 87 (21.80) |
| N2 | 99 (24.81) |
| N3 | 53 (13.28) |
| *TNM Stage* | |
| Stage I | 68 (17.04) |
| Stage II | 67 (16.79) |
| Stage III | 105 (26.32) |
| Stage IV | 159 (39.85) |
| *Stage ItmIV* | |
| Stage 0 | 1 (0.25) |
| Stage I | 66 (16.54) |
| Stage II | 66 (16.54) |
| Stage III | 98 (24.56) |
| Stage IV | 159 (39.85) |
| Missing | 9 (2.26) |

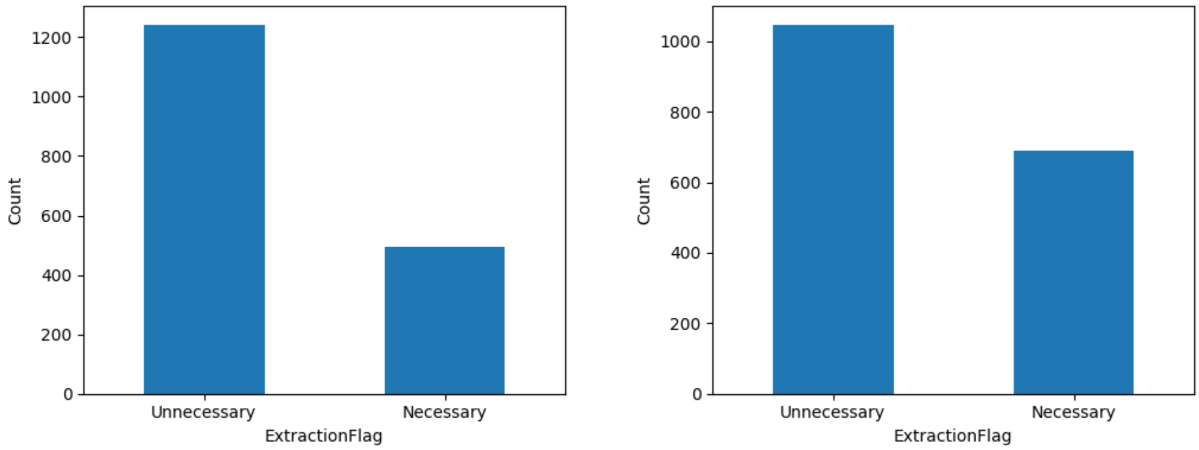
| **Variable** | **n (%)** |
|---|---|
| *Tumor Category* | |
| Type 1 | 325 (81.45) |
| Type 2 | 36 (9.02) |
| Type 3 | 2 (0.50) |
| Type 4 | 22 (5.51) |
| Type 5 | 2 (0.50) |
| Type 6 | 11 (2.76) |
| Missing | 1 (0.25) |
| *Tumor Site* | |
| Site 1 | 51 (12.78) |
| Site 2 | 16 (4.01) |
| Site 3 | 121 (30.33) |
| Site 4 | 40 (10.03) |
| Site 5 | 69 (17.29) |
| Site 6 | 23 (5.76) |
| Site 7 | 10 (2.51) |
| Site 8 | 63 (15.79) |
| Site 9 | 5 (1.25) |
| Missing | 1 (0.25) |
| *Stage* | |
| Stage 0 | 1 (0.25) |
| Stage 1 | 66 (16.54) |
| Stage 2 | 66 (16.54) |
| Stage 3 | 98 (24.56) |
| Stage 4 | 105 (26.32) |
| Stage 5 | 52 (13.03) |
| Stage 6 | 1 (0.25) |
| Missing | 10 (2.51) |
| *Systemic Therapy* | |
| Therapy 0 | 107 (26.82) |
| Therapy 1 | 28 (7.02) |
| Therapy 2 | 2 (0.50) |
| Therapy 4 | 256 (64.16) |
| Therapy 5 | 4 (1.00) |
| Therapy 6 | 1 (0.25) |
| Missing | 1 (0.25) |

lying between 57 and 71 years. Furthermore, TotalDose, reflecting the prescribed RT dose to the tumor site rather than individual teeth, averaged 67.08 Gy. The table indicates that the majority of patients (25–75%) received relatively high total doses, ranging from 66 to 70 Gy. It also shows that on average, 5.54 teeth were extracted per patient. However, when applying the 40 Gy dose threshold, on average, only 1.26 extractions per patient were necessary according to $D_{\mathrm{mean}}$, and 1.75 per patient according to $D_{\mathrm{max}}$.

Table 4: Descriptive statistics for continuous variables ($n$ = 399) in the Tooth Extraction Dataset

| **Variable** | **Mean** | **Std** | **Min** | **25%** | **50%** | **75%** | **Max** |
|---|---|---|---|---|---|---|---|
| Age | 63.62 | 11.32 | 22.0 | 57.0 | 64.0 | 71.0 | 93.0 |
| TotalDose (Gy) | 67.08 | 5.20 | 36.0 | 66.0 | 68.0 | 70.0 | 72.0 |
| nExtracted | 5.54 | 5.11 | 1.0 | 2.0 | 4.0 | 7.5 | 31.0 |
| nJustified_mean | 1.26 | 2.26 | 0.0 | 0.0 | 0.0 | 2.0 | 14.0 |
| nUnjustified_mean | 3.23 | 4.08 | 0.0 | 1.0 | 2.0 | 4.0 | 28.0 |
| nJustified_max | 1.75 | 2.63 | 0.0 | 0.0 | 1.0 | 2.0 | 14.0 |
| nUnjustified_max | 2.73 | 3.82 | 0.0 | 0.0 | 2.0 | 3.0 | 28.0 |

To explore this further, the total number of extracted teeth, among all patients, was plotted against each tooth's radiation exposure relative to the 40Gy risk threshold. As illustrated in Figure 6, only 28% of extracted teeth had a $\mathrm{D}_{\mathrm{mean}} \geq 40$ Gy, and 39% had a $\mathrm{D}_{\mathrm{max}} \geq 40$ Gy, levels at which extraction would have been clinically justified. However, the graphs show that the majority of extractions occurred below this threshold, with 61% by $\mathrm{D}_{\mathrm{max}}$ and 72% by $\mathrm{D}_{\mathrm{mean}}$ receiving less than 40 Gy. These findings reveal a significant imbalance in the distribution of extractions relative to the threshold, with the majority of teeth being extracted at lower radiation doses. This imbalance highlights the need for careful modeling to address potential bias and to properly account for the underrepresented high-dose cases.




(a) $D_{\mathrm{mean}}$ (b) $D_{\mathrm{max}}$

Figure 6: Number of extractions categorized as clinically necessary, using (a) the $D_{\mathrm{mean}}$ and (b) the $D_{\mathrm{mean}}$ as reference metric

Furthermore, a thorough exploratory analysis was conducted to investigate whether certain clinical variables were associated with higher rates of necessary or unnecessary extractions. Categorical variables such as TumorLaterality, TNM_T, TNM_N, TNM_Stage, TumorCategory, StageItmIV, and SystemicTherapy generally exhibited a consistent pattern across all levels, with a higher number of unnecessary extractions compared to necessary ones. No specific level within these variables stood out as an exception to this trend.

However, the variable TumorLocation revealed interesting differences. As shown in Figure 7, certain tumor locations were associated with different rates of necessary versus unnecessary extractions, suggesting that TumorLocation may be an important factor influencing extraction decisions. Tumor Location 4 had the highest rate of unnecessary extractions for both $D_{\mathrm{mean}}$ and $D_{\mathrm{max}}$, showing a clear imbalance. Location 3 followed a similar trend, with most extractions being unnecessary. Locations 7 and 8 also stood out, while they had fewer extractions overall, unnecessary extractions still outweighed the necessary ones for both dose metrics. In contrast, Location 1 was the only area where necessary extractions were more common than unnecessary ones under $D_{\mathrm{mean}}$ and $D_{\mathrm{max}}$, implying teeth here were more likely to receive clinically significant radiation doses.

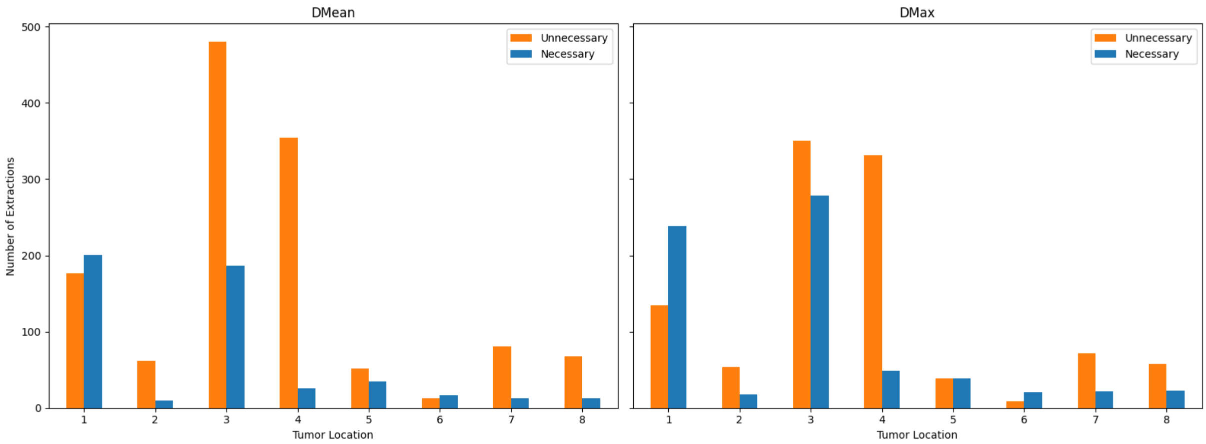



Figure 7: Distribution of extraction decisions by TumorLocation, classified with a 40Gy threshold based on (left) $D_{\text{mean}}$ and (right) $D_{\text{max}}$

A similar pattern was observed for the variable TumorSite, as shown in Figure 8. The distribution of necessary versus unnecessary extractions closely resembled that of TumorLocation, with Sites 3, 4, 5, and 8 again showing disproportionately high rates of unnecessary extractions across both $D_{\text{mean}}$ and $D_{\text{max}}$. Likewise, Site 1 stood out with a higher proportion of necessary extractions, consistent with earlier findings.

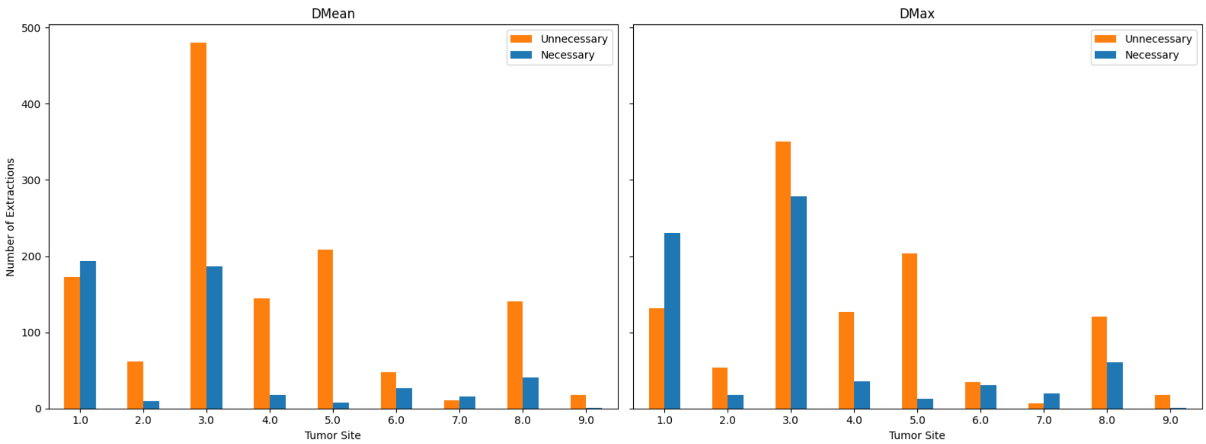



Figure 8: Distribution of extraction decisions by TumorSite, classified with a 40Gy threshold based on (left) $D_{\text{mean}}$ and (right) $D_{\text{max}}$

Extractions were also evaluated per ToothType based on whether they were classified as necessary or unnecessary according to the 40 Gy threshold. As shown in Figure 9, the majority of extractions across tooth types were unnecessary. However, under both $D_{\text{mean}}$ and $D_{\text{max}}$, teeth 36, 37, and 38 stood out as exceptions, with more necessary than unnecessary extractions. Additionally, under $D_{max}$, tooth 35 also showed a higher number of necessary extractions. Hence the figure shows tooth location matters as some spots are far more likely to receive higher radiation doses than others.

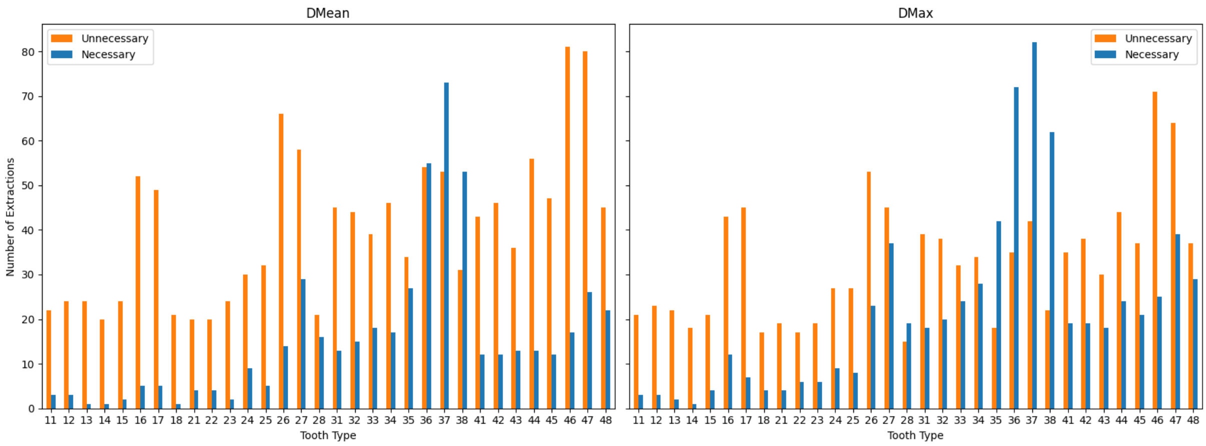



Figure 9: Distribution of extraction decisions by ToothType, classified with a 40Gy threshold based on (left) $D_{mean}$ and (right) $D_{max}$

## 4.2 Data Preparation

Given the limited number of variables available at dental screening, all non-demographic features were included as predictors for the ML models. The demographic variables age and sex were excluded to reduce the risk of introducing bias into predictions. The resulting predictor set comprises ToothType, TumorLaterality, TumorLocation, TNM_T, TNM_N, TNM_Stage, TotalDose, TumorCategory, TumorSite, Stage, StageIt-mIV and SystemicTherapy.

For modeling, the dataset was reorganized from the patient level to the tooth level. In the tooth-level format, each row represents a single tooth and is supplemented with the corresponding patient's clinical variables as well as the variable ToothType, indicating which tooth is under consideration and whether it was extracted. Two datasets were created by defining the outcome variable $y$ using one of two extraction flags. First, the $D_{mean}$ dataset where $y$ = *ExtractionFlag_dmean*, and the $D_{max}$ dataset where $y$ = *ExtractionFlag_dmax*.

As only a small number of variables contained missing values, cases with incomplete data were excluded from the analysis. Both resulting datasets had a shape of (1489, 15). Each dataset was split into training (80%) and test (20%) subsets at the patient level, ensuring that all teeth from the same patient were assigned to either the training or test set, but not both. For the LR, SVM with both linear and rbf kernels, and RF models, categorical variables were encoded using OHE. For the SVM models that employed a categorical kernel, OHE was not necessary, as the kernel is designed to handle categorical variables directly.

### 4.3 Additional Datasets for Kernel Experiments

In addition to the main clinical dataset, two supplementary datasets were retrieved from UCI Machine Learning Repository (2025). These are employed to evaluate the performance of the extension of the categorical kernel. The first is the Congressional Voting dataset, comprising 435 instances with 16 categorical vote features and a binary target. The second is the MONK's Problems dataset, comprising 436 instances and six categorical features with 2 to 4 levels. Both datasets were complete and required no preprocessing prior to modeling. Both supplementary datasets employed the same encoding scheme as described in Section 4.2, but used a random 80/20 split for training-test partitioning.

## 5 Model Configuration and Experimental Plans

This section documents the implementation of all models and analyses performed. It first covers the baseline models, detailing the datasets, and SHAP interpretability setup. It then outlines the SVM framework, including the standard and categorical kernels and their use in ensemble methods. Lastly, it describes the implementation of the CAE as a substitute for OHE.

All experiments are conducted on both the $D_{mean}$ and $D_{max}$ datasets, with models predicting the probability of Class 1. This probability is interpreted as the estimated risk of requiring extraction, whereas Class 0 corresponds to cases where no extraction is necessary. A threshold of 50% is applied for classification, with additional analyses evaluating performance across the full probability range.

## 5.1 Baseline Models

To establish a benchmark, three classifiers (LR, SVM, RF) were constructed to evaluate how effectively standard ML methods can predict tooth extraction decisions from the original set of clinical features. These models were trained on both the $D_{mean}$ and $D_{max}$ datasets using OHE features. The SVM and RF were wrapped in CalibrationCV with Platt scaling to produce calibrated probability estimates. The baseline models are further examined through calibration plots and prediction histograms. These analyses provide deeper insight into the reliability of probability estimates.

Finally, model behavior was interpreted by analyzing feature contributions via Shapley values, a framework from cooperative game theory adapted for feature attribution in ML (Rozemberczki et al. 2022). Given a model $f$ and a set of input features $X = \{x_1, x_2, \ldots, x_d\}$, the Shapley value $\phi_i$ for feature $x_i$ is defined as the average marginal contribution of $x_i$ to the model's prediction over all possible subsets of features:

$$\phi_i = \sum_{S \subseteq X \setminus \{x_i\}} \frac{|S|!\,(d-|S|-1)!}{d!} \Big[ f\big(S \cup \{x_i\}\big) - f(S) \Big].$$

Intuitively, $\phi_i$ quantifies how much adding feature $x_i$ to a coalition $S$ of other features changes the prediction, averaged across all coalitions.

The implementation is performed using the SHAP Python package, with LinearExplainer for linear models and TreeExplainer for tree-based models. For the SVM explanations, KernelExplainer was applied on a subsample given its computational cost with 88 features. Therefore, 50 randomly selected training instances were used as the background set and 30 test instances for explanation, drawn from the full dataset. For each model, the absolute Shapley values are averaged over the validation set to produce a global importance score:

$$\text{Importance}(x_i) = \frac{1}{N} \sum_{j=1}^{N} \big|\phi_i^{(j)}\big|.$$

## 5.2 Support Vector Machines Kernel Comparison

The second experiment evaluates the performance of the proposed $k_2$ kernel against the original $k_1$ kernel by Belanche and Villegas (2013) and standard baselines. This comparison is conducted by building SVM models with four different kernels, namely, the linear, rbf, $k_1$, and the proposed $k_2$ kernel. These include the the core $D_{mean}$ and $D_{max}$ medical datasets, as well as the Congressional Voting and MONK's Problems datasets to validate performance on established categorical data benchmarks.

On the medical datasets, the model predics the probability of Class 1. On the benchmark datasets, standard classification accuracy is used. Because CalibrationCV does not support custom kernels, all models use the SVM's native probability estimation for a consistent comparison.

## 5.3 Ensemble Support Vector Machines

This study also investigates the integration of categorical kernels into SVM ensemble methods, an approach previously unexplored in the literature. Ensembles were constructed using the RSM and RSVM methods on three datasets. These include the $D_{mean}$, Congressional Voting, and MONK's Problems dataset. Each ensemble was built using three kernel types, namely, the linear, rbf, and the $k_1$ kernel by Belanche and Villegas (2013).

To analyze performance, both ensemble size $|E|$ and the number of features $m$ sampled per base learner were systematically varied, with the maximum $|E|$ capped at 50 due to computational constraints from hyperparameter tuning. Classification accuracy was evaluated as a function of $|E|$ and $m$ for each kernel type. The linear and rbf kernels used OHE for categorical variables, while the $k_1$ kernel operated directly on raw categorical features. Table 5 summarizes the feature counts $m$ used in these experiments.

Table 5: Number of features used per dataset and kernel in RSM and RSVM models

| **Dataset** | **Kernel** | **Feature count ($m$)** |
|---|---|---|
| $D_{mean}$ | linear, rbf | 30, 40, 50, 60, 70, 80, 88 |
| | $k_1$ | 4, 6, 8, 10, 12 |
| Synthetic Poisson | linear, rbf | 10, 15, 30, 35, 39 |
| | $k_1$ | 3, 4, 6 |
| Congressional Voting | linear, rbf | 10, 18, 25, 28, 32 |
| | $k_1$ | 36, 9, 12, 16 |

### 5.4 Categorical Autoencoder

The final experiment evaluates a CAE as an alternative preprocessing method for categorical features. The CAE is trained separately on the $D_{mean}$ and $D_{max}$ datasets to generate a low-dimensional representation of the original categorical features. These compressed representations then replace the OHE features as input to an SVM with an rbf kernel.

To assess the effectiveness of this approach, the CAE-SVM model is compared against a baseline SVM with an rbf kernel trained directly on the OHE features. The architectural details and optimal hyperparameters for the CAE are provided in Appendix E (Table 17).

## 6 Results and Discussion

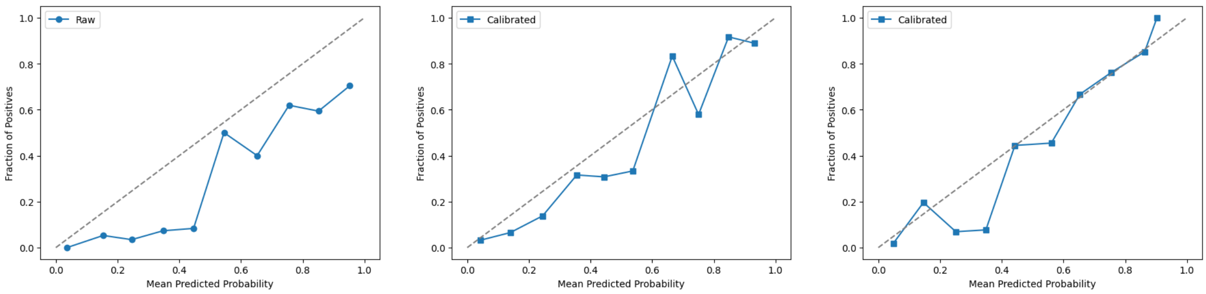
### 6.1 Baseline Models

Table 6 shows the performance metrics of the baseline models, where the decision $y$ is driven by the $D_{mean}$ metric and by applying the default 0.5 probability cutoff. LR appears to have the worst perfomance compared to both SVM and RF, with the highest BS (0.1539) and bCE (0.4327) and the lowest overall accuracy (0.80). In contrast, SVM and RF are have similar perfomance in terms of the BS, bCE and accuracy. For Class 0, both SVM and RF achieve very high precision and recall ($>$ 0.91), whereas LR yields lower recall (0.77). For Class 1, LR actually leads on recall (0.90) but at the expense of precision (0.56). For Class 1, SVM (0.73) and RF (0.71) achieve recall values that are noticeably lower than LR. However, both maintain precision above 0.7, indicating fewer false positives. These results are based on the standard 0.5 probability cutoff, which may not represent the optimal threshold.

Table 6: Performance of calibrated models predicting extraction necessity based on $D_{mean}$ dataset and threshold 0.5

| **Model** | **BS** | **bCE** | **Acc** | **Class 0** | | | **Class 1** | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | P | R | $F_1$ | P | R | $F_1$ |
| LR | 0.1539 | 0.4327 | 0.80 | 0.96 | 0.77 | 0.86 | 0.56 | 0.90 | 0.69 |
| SVM | 0.0971 | 0.4257 | 0.87 | 0.92 | 0.91 | 0.91 | 0.72 | 0.73 | 0.73 |
| RF | 0.0948 | 0.4194 | 0.88 | 0.91 | 0.93 | 0.92 | 0.76 | 0.71 | 0.73 |

Figure 10 compares the calibration curves of the three classifiers. Perfect calibration would align all points along the dashed 45° line, where points below the line indicate over-confidence, while points above the line indicate under-confidence. The LR model consistently overestimates the risk of tooth extraction, particularly in the 0–0.4 and 0.6–1.0 probability ranges. In contrast, the SVM demonstrates improved reliability. Predictions up to 0.6 remain slightly overconfident, but beyond this point the curve follows the diagonal more closely, indicating well-calibrated estimates for Class 1 cases. Lastly, the RF exhibits minor overconfidence in the 0.2–0.4 range but otherwise it tracks the diagonal closely, achieving near perfect alignment, where extraction is most probable.




(a) LR (b) SVM (c) RF

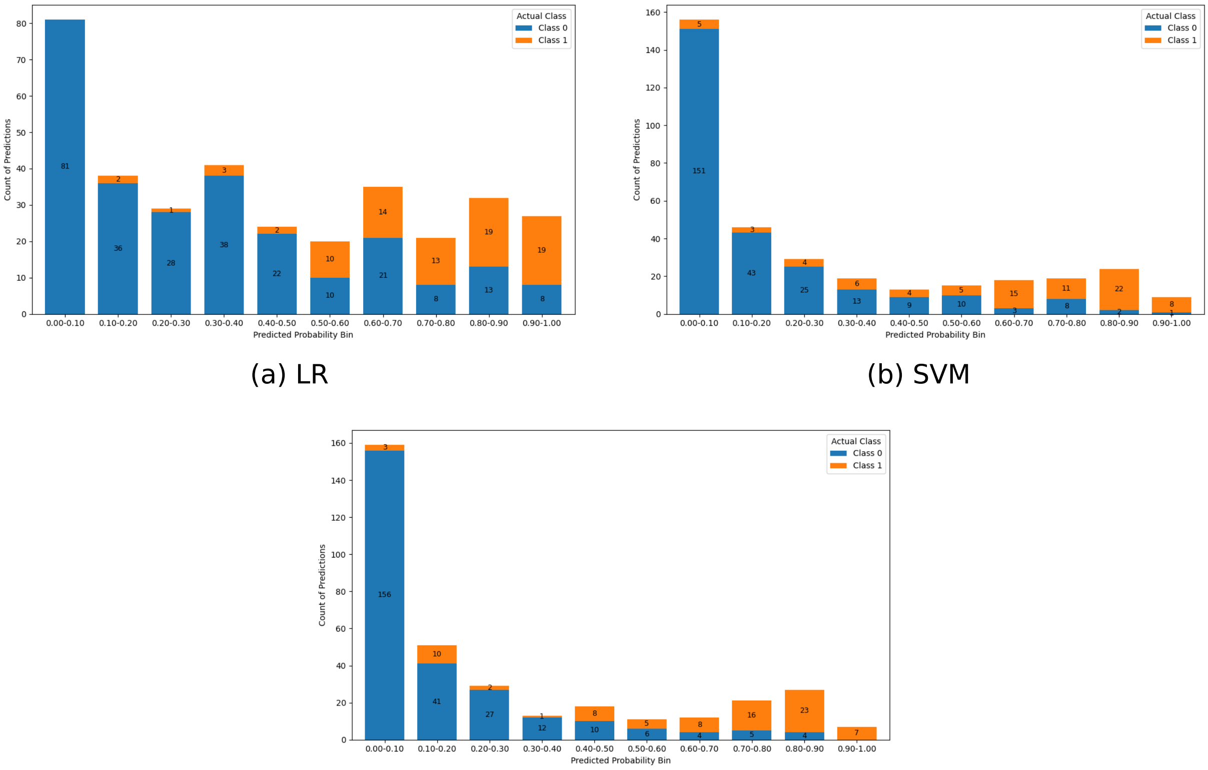
Figure 10: Calibration curves of the baseline models on $D_{mean}$ dataset

Figure 11 shows the distribution of predicted probabilities of Class 1 cases, binned in 0.1 intervals and stacked by true outcome. The plots reveal differences in how each model distinguishes between Class 0 and Class 1.

The LR concentrates Class 0 cases in low-probability bins (<0.5), with few FN. For instance, only 8 misclassified in the 0–0.5 region. However, it shows significant overlap between classes in the 0.5–1.0 range for Class 1. SVM has a more extreme separation of Class 0, with nearly all negative instances in the 0.0–0.1 bin correctly assigned to the lowest probability bins. However, this comes at the cost of increased misclassification of Class 1 cases in the $< 0.5$ range compared to LR. The model's separation improves at $\geq 0.6$, where Class 1 dominates and there are fewer Class 0 bins in these bins compared to LR, which minimizes the number of unnecessary extracted teeth. Lastly, RF demonstrates a similar pattern to SVM, concentrating most Class 0 instances in the 0.0–0.1 bin. However, this is also linked to more misclassifications of Class 1 in low-probability bins compared to the LR model. Furthermore, RF outperforms both LR and SVM in high-confidence predictions ($\geq 0.7$), where Class 1 representation is strongest. Lastly, both SVM and RF, exhibit poor class separation in mid-range probabilities (0.4–0.6), reflecting greater classification uncertainty.

The calibration plots (Figure 10) and probability histograms (Figure 11) reveal the clinically important trade-offs between models. While LR minimizes FNs at low probabilities ($< 0.5$), it generates significant class overlap in higher ranges (0.5–1.0), leading to excessive unnecessary extractions and consistently overestimating the probability of Class 1. In contrast, SVM and RF demonstrate strong performance for this task, achieving high class-specific recall at the upper and lower ends of the probability scale. However, their performance is less reliable in the intermediate range (0.2–0.6). This pattern suggests that a clinical decision-support framework could be developed where high- and low-confidence predictions could be automated, while intermediate cases require clinician review, though these thresholds should be confirmed using bootstrapping for clinical implementation.

Table 7 shows the performance metrics of the baseline models, where the decision $y$ is driven by the $D_{max}$ metric. LR again exhibits the weakest overall performance (BS: 0.1676, bCE: 0.52, accuracy: 0.76) but achieves the highest recall for the Class 1 (0.84), surpassing both SVM (0.80) and RF (0.74). The LR exhibits the



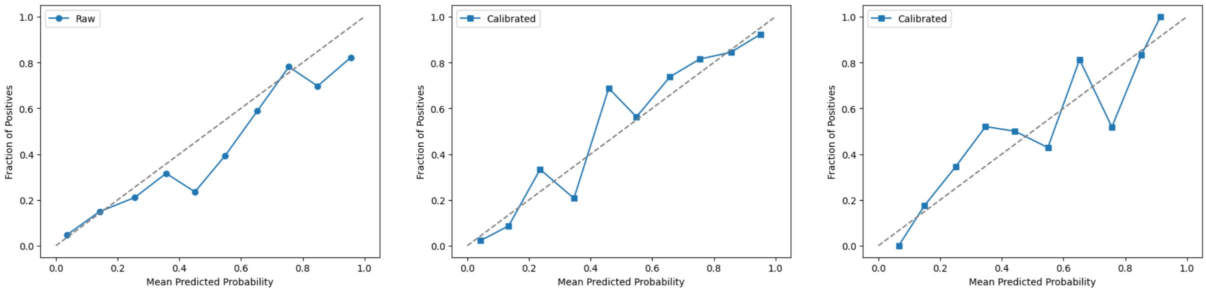

(a) LR

(b) SVM

(c) RF

Figure 11: Distribution of predicted probabilities of the baseline models on $D_{mean}$ dataset

same behavior as under the classification driven by the $D_{mean}$ metric, it maximizes Class 1 recall at the expense of precision. SVM appears the have the best balance. It maintains strong recall (0.80) while achieving the highest precision (0.82) and F1 score (0.81) for Class 1. Its recall is only slightly lower than LR's (0.80 vs. 0.84), but SVMs precision makes it more reliable for practical use. Lastly, RF falls behind both LR and SVM in terms of the recall of Class 1, which suggests RF fails to generalize as effectively as SVM.




(a) LR (b) SVM (c) RF

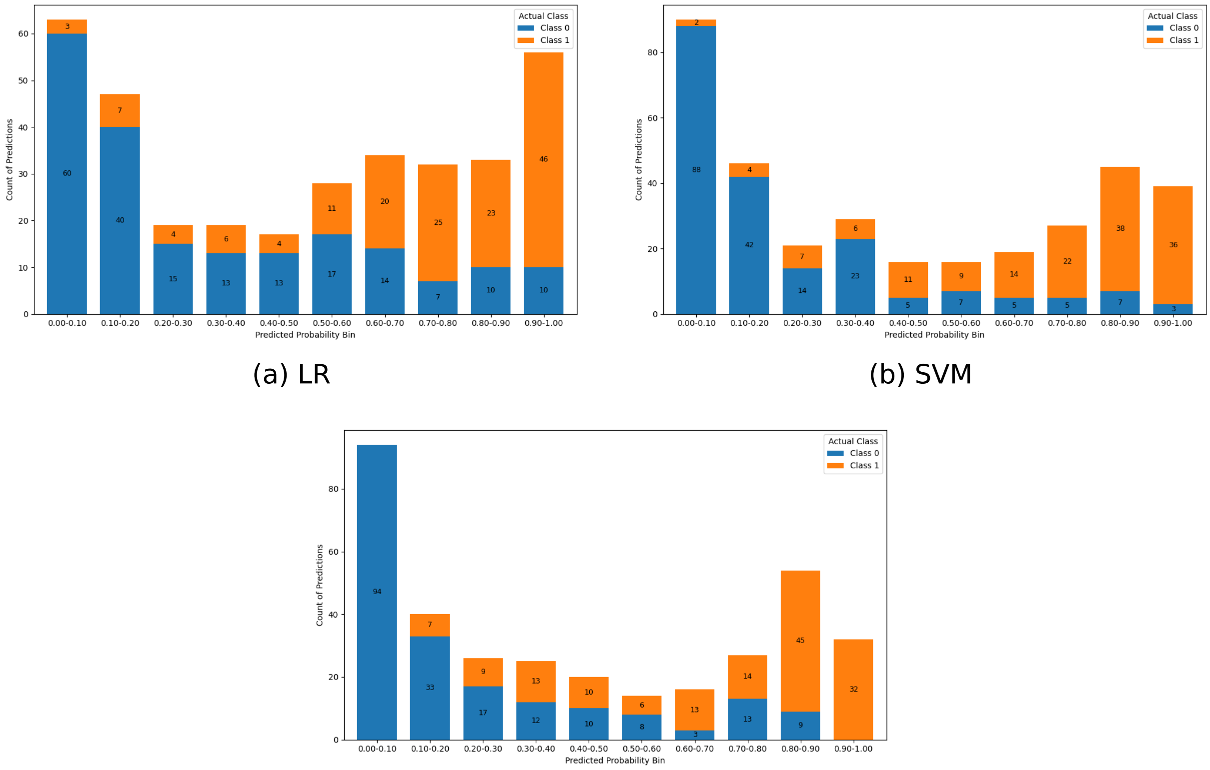
Figure 12: Calibration curves of the baseline models on $D_{max}$ dataset

Table 7: Performance of calibrated models predicting extraction necessity based on $D_{max}$ and threshold 0.5

| **Model** | **BS** | **bCE** | **Acc** | **Class 0** | | | **Class 1** | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | P | R | $F_1$ | P | R | $F_1$ |
| LR | 0.1676 | 0.5194 | 0.76 | 0.85 | 0.71 | 0.77 | 0.68 | 0.84 | 0.75 |
| SVM | 0.1156 | 0.3823 | 0.84 | 0.85 | 0.86 | 0.86 | 0.82 | 0.80 | 0.81 |
| RF | 0.1330 | 0.4155 | 0.79 | 0.81 | 0.83 | 0.82 | 0.77 | 0.74 | 0.75 |

Figure 12 compares the calibration curves of the three models. LR closely follows the ideal diagonal, indicating good calibration. It shows only a slight underconfidence in the mid-range (0.4–0.6), where it modestly underestimates the true risk. The calibrated SVM also follows the ideal diagonal line very closely, also with only a minor underconfidence in the mid-range (0.4–0.6). Lastly, the calibrated RF exhibits the greatest miscalibration. It is overconfident for predictions in the 0.3–0.5 range and underconfident at the higher end (0.7-0.8).

Figure 12 shows the distribution of predicted probabilities for Class 1 according to the $D_{max}$. For the LR model, most Class 0 instances are correctly assigned very low probabilities (0.0–0.2), however, there are some Class 1 instances that fall into this range. In the intermediate range (0.5-0.7), the model demonstrates substantial overlap between the two classes. The Class 1 instances predominate above the 0.6 range. Overall, LR assigns a significant number of Class 0 to probabilities exceeding the 0.5 decision threshold. In contrast, the SVM appears to be more decisive. Nearly




(a) LR (b) SVM

(c) RF

Figure 13: Distribution of predicted probabilities of the baseline models on $D_{max}$ dataset

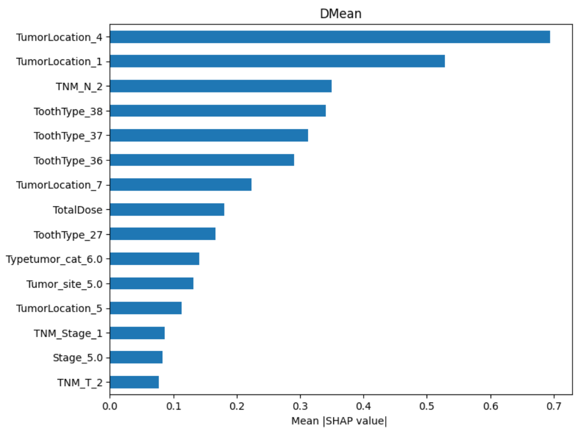
all Class 0 instances concentrate below 0.2, while most Class 1 predictions are packed above 0.8. In addition, less instances fall in the middle range (0.3–0.7), meaning the SVM strongly pushes probabilities toward the extremes. In comparison to LR, it has fewer uncertain Class 0 predictions above 0.5, reducing FP. Likewise, RF strongly clusters Class 0 predictions below 0.2 and Class 1 predictions above 0.8, but slightly has higher fraction of negatives to exceed the 0.5 threshold. RF also assigns more TPs to probabilities below 0.5 than either LR or SVM, resulting in a higher FN rate in that region.

For clinical implementation based on the $D_{max}$ analysis, both SVM and RF models yield reliable predictions at the extremes of the probability scale, but their performance becomes notably less consistent in the mid-range (0.2–0.8), where predictive confidence diminishes. The observed thresholds offer preliminary guidance for a potential decision-support system, though as previously mentioned, their reliability must be verified through bootstrapping.

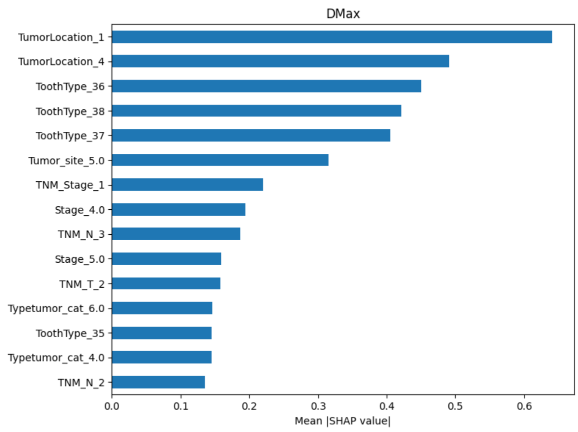
## 6.2 Feature Importance on the Dental Extraction Dataset

Figures 14, 15, and 16 present the Shapley (SHAP) feature importance plots for each of the baseline models. Each figure contains two subplots corresponding to the two radiation dose metrics used. The left subplot in each figure shows feature importances with respect to the $D_{mean}$, while the second on the right shows feature importances with respect to the $D_{max}$. Across all three classifiers and for both $D_{mean}$ and $D_{max}$, the categorical indicators for TumorLocation, ToothType dummies and TotalDose emerge as important predictors.

For the LR model (See Figure 14), the $D_{mean}$ SHAP plot is dominated by TumorLocation dummies. TumorLocation_4 and TumorLocation_1 have the largest average impacts, followed by the nodal-stage indicator TNM_N_2 and tooth-type flags (38, 37, 36). Total prescribed dose and the lower-order tumor category sit in the middle of the ranking, while overall clinical stage and T-stage (Stage_5.0, TNM_T_2) contribute very little. In the maximum-dose ($D\_max$) setting, TumorLocation_1 slightly overtakes location 4, and tooth types remain top predictors.



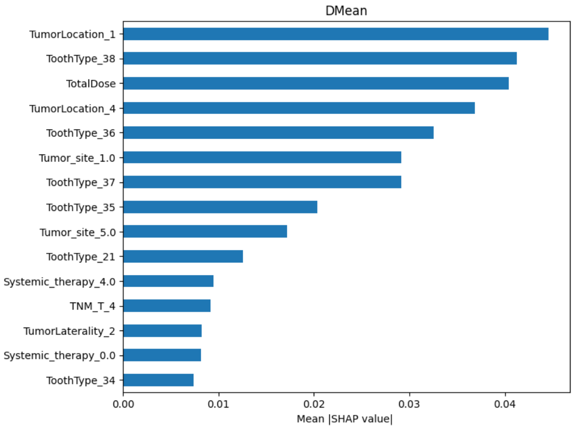

(a) SHAP values based on $D_{mean}$



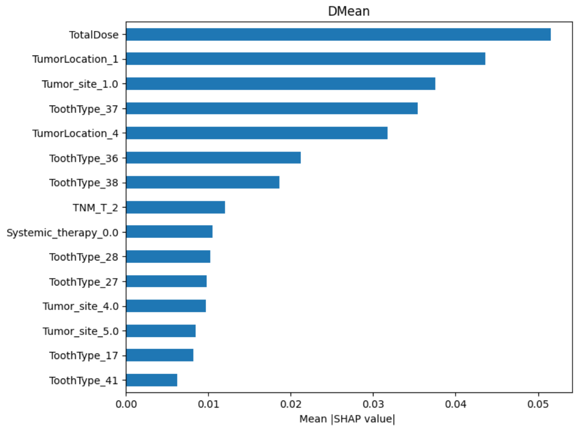

(b) SHAP values based on $D_{max}$

Figure 14: SHAP feature importance of LR classifier based on $D_{mean}$ (a) and $D_{max}$ (b)

For the SVM model (Figure 15), the $D_{mean}$ is similarly dominated by TumorLocation dummy variables. TumorLocation_1 emerges as the most influential feature, followed by ToothType_38 and the TotalDose. TumorLocation_4 also ranks highly, alongside other tooth-type indicators (36, 37, 35). In the $D_{max}$ setting, the plot indicates that TotalDose is the most important feature followed by TumorLocation_1, Tumor_site_1, ToothType_37, TumorLocation_4.



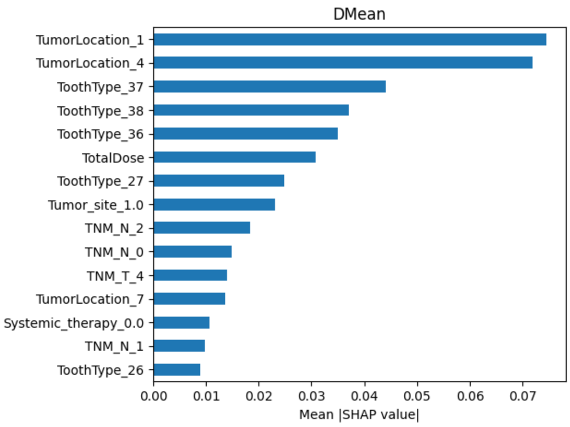

(a) SHAP values based on $D_{mean}$



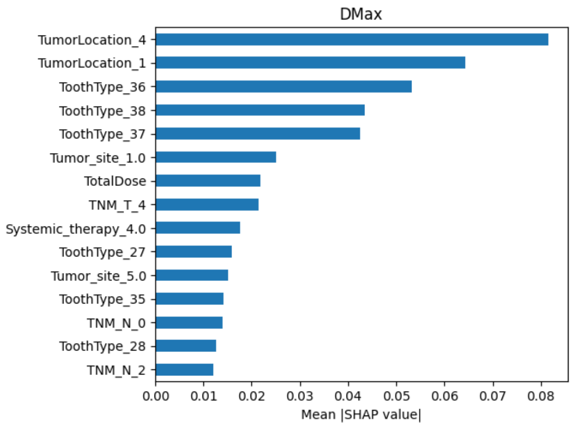

(b) SHAP values based on $D_{max}$

Figure 15: SHAP feature importance of SVM classifier based on $D_{mean}$ (a) and $D_{max}$ (b)

For the RF model (See Figure 16), the $D_{mean}$ SHAP plot still places TumorLocation_1 and TumorLocation_4 at the top, again, followed by ToothType flags (37, 38, 37). Here, total prescribed dose, TNM indicators and TumorSite flag surfaces in the mid-rankings. In the $D_{max}$ view, TumorLocation_4 leads, followed by location 1. ToothType, TotalDose and TNM indicators remain influential.




(a) SHAP values based on $D_{mean}$




(b) SHAP values based on $D_{max}$

Figure 16: SHAP feature importance for RF classifier based on $D_{mean}$ (a) and $D_{max}$ (b)

Across all models and for both dose settings, TumorLocation and ToothType indicators as well as the TotalDose consistently rank among the top predictors for extraction risk in HNC patients before RT. Although the importance ordering of those features can shift slightly between the $D_{mean}$ and $D_{max}$ settings, these variables almost always occupy the leading positions, with SHAP values gradually tapering off thereafter.

## *6.3 Categorical-Kernel Support Vector Machine*

Table 8 presents the optimal hyperparameters for the SVM using the original categorical kernel $k_1$ and the adjusted kernel $k_2$. It shows that the optimal value for the parameter $\sigma$ is consistently 0.001 across all datasets. This setting effectively forces the weighting function $g$ in $k_2$ to be close to zero, thereby making $k_2$ nearly identical to $k_1$. This outcome demonstrates that the SVM does not benefit from assigning a nonzero similarity score to pairs of distinct rare categories.

Table 8: Best hyperparameters for categorical kernels $k_1$ and $k_2$, in SVM

| **Dataset** | $k_1$ | | | $k_2$ | | | |
|---|---|---|---|---|---|---|---|
| | $\gamma$ | $\alpha$ | $C$ | $\gamma$ | $\alpha$ | $C$ | $\sigma$ |
| $D_{mean}$ | 1 | 1.5 | 10 | 1 | 1.5 | 10 | 0.001 |
| $D_{max}$ | 4 | 0.7 | 1 | 4 | 0.9 | 1 | 0.001 |
| MONK's Problem | 4 | 0.7 | 0.1 | 4 | 0.7 | 0.1 | 0.001 |
| Congressional Voting | 1 | 1.0 | 10 | 1 | 1.5 | 10 | 0.001 |

Table 9 and Table 10 show the performance of the SVM models using the three distinct kernels (rbf, $k_1$, $k_2$). Across both the $D_{mean}$ and $D_{max}$ experiments, all three kernels yielded very similar BS and bCE values, with comparable overall accuracy. In the $D_{mean}$ scenario, the custom kernels $k_1$ and $k_2$ achieved higher recall on Class 1 than the rbf kernel. In the $D_{max}$ setting, however, there were no meaningful differences across kernels in any of the class-specific metrics, P, R and F1 score, were effectively equivalent for both Class 0 and Class 1.

Table 9: Performance of SVM models predicting extraction necessity based on $D_{mean}$ and threshold 0.5

| **Kernel** | **BS** | **bCE** | **Acc** | **Class 0** | | | **Class 1** | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | P | R | $F_1$ | P | R | $F_1$ |
| rbf | 0.0971 | 0.4257 | 0.87 | 0.92 | 0.91 | 0.91 | 0.72 | 0.73 | 0.73 |
| k1 | 0.0960 | 0.3984 | 0.88 | 0.93 | 0.90 | 0.92 | 0.72 | 0.80 | 0.75 |
| k2 | 0.0969 | 0.3962 | 0.87 | 0.93 | 0.89 | 0.91 | 0.70 | 0.80 | 0.75 |

Table 10: Performance of SVM models predicting extraction necessity based on $D_{max}$ and threshold 0.5

| **Kernel** | **BS** | **bCE** | **Acc** | **Class 0** | | | **Class 1** | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | P | R | $F_1$ | P | R | $F_1$ |
| rbf | 0.1156 | 0.3823 | 0.84 | 0.85 | 0.86 | 0.86 | 0.82 | 0.80 | 0.81 |
| k1 | 0.1281 | 0.4119 | 0.82 | 0.85 | 0.83 | 0.84 | 0.78 | 0.81 | 0.80 |
| k2 | 0.1266 | 0.4133 | 0.82 | 0.85 | 0.83 | 0.84 | 0.78 | 0.81 | 0.79 |

Appendix G displays the SVM performance with the three kernels on two additional datasets. The results for the MONK's Problems dataset are shown in Table 18, and those for the Congressional Voting Records are in Table 19. Standard classification was applied, and all kernels achieved comparable accuracy, precision, recall and F1-score. No meaningful differences emerged across kernels for any class-specific metric in either Class 0 or Class 1. These findings demonstrate that categorical kernels offer an effective alternative to OHE, as they perform comparably to standard continuous kernels while operating directly on raw categorical data. However, $k_2$ provides no performance benefit over $k_1$, as the off-diagonal similarity values assigned to distinct categories do not lead to improved classification on these datasets.

Furthermore, Figures 20–22 in Appendix G present two-dimensional Kernel PCA projections for SVMs employing the Hamming, $k_1$, $k_2$, and rbf kernels. For the $D_{mean}$ dataset (See Figure 20), all kernels produced clusters with substantial class overlap. The categorical $k_2$ kernel yielded the tightest clusters compared to the other kernels. On the Congressional Voting dataset (See Figure 21), a pronounced arc-shaped pattern emerged for the $k_1$ and rbf kernels, within which a clear separation of the two classes is visible. The Hamming kernel also achieved clear separation, however, the data points are more spread across the feature space. The $k_2$ projection appears more spread out than $k_1$, still slightly following the arc structure. Finally, in the MONK's Problems dataset (See Figure 22), the Kernel PCA projections are characterized by extremely tight, overlapping clusters. Among them, the clusters produced by the $k_2$ kernel are the most condensed.

## 6.4 Ensemble Support Vector Machines

Ensemble SVMs with the $k_1$ kernel were evaluated by varying the ensemble size $|E|$ and the feature-subset size $m$ under both RSM and RSVM frameworks. Performance is summarized by plotting test accuracy across $|E|$ and $m$. Figures 17 and 18 report test accuracy on the $D_{mean}$ dataset for the linear, rbf, and $k_1$ kernels. Additional results on the MONK's Problems dataset as well as the Congressional Voting Dataset can be found in Appendix H.


Figure 17: Performance of SVM ensembles with linear and rbf kernels across different numbers of features ($m$) and ensemble sizes on $D_{mean}$ dataset

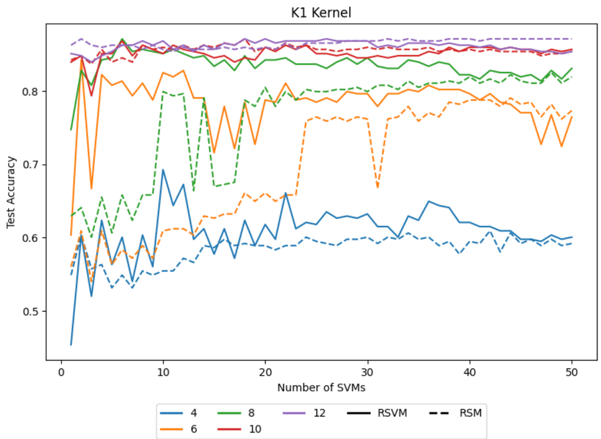



Figure 18: Performance of SVM ensemble with $k_1$ kernel across different numbers of features ($m$) and ensemble sizes on $D_{mean}$ dataset

The plots show that all three kernels exhibit rapid accuracy gains over the first 5 to 10 base learners, followed by a plateau after 20 to 30 models, with the RSM and RSVM having similar learning curves. The largest performance gap between RSM an RSVM appears when $m$ is smallest (see Appendix H). The $k_1$ ensemble consistently lags behind its linear and rbf counterparts, especially when the feature-subset size $m$ is small. On the $D_{mean}$ dataset, the linear and rbf ensembles achieve higher plateau accuracies than $k_1$ ensemble. Only on the Congressional Voting dataset (See Figure 23 & 24) do all three kernels converge to similar performance once each ensemble saturates. The results of the MONK's Problems dataset (See Figure 25 & 26) show that $k_1$ and rbf ensembles performed better than linear. Interestingly, the rbf kernel could achieve perfect accuracy even when individual ensemble members use a small subset of features (e.g., 7 out of 17). In contrast, the $k_1$ kernel only reached perfect accuracy when each member used the complete feature set, effectively reducing the ensemble to a single model.

The performance of the ensemble methods was also compared to their standalone kernel counterparts. The analysis reveals that a standalone SVM consistently matched or surpassed ensemble performance. For the $D_{mean}$ dataset (see Table 9), a single SVM, using either rbf or $k_1$ kernel, achieved an accuracy of approximately 0.88, outperforming all ensemble variants. This also held for the Congressional Voting Records (see Table 19), where the high accuracy of the standalone kernels (0.95–0.97) was a limit that the ensembles could not surpass. The only exception was the MONK's Problems dataset (see Table 18), where ensembles matched the perfect accuracy of the standalone SVM. However, this still provided no performance gain. Consequently, the ensemble methods, implementing categorical kernel, offered no advantage over a well-tuned, standalone SVM across the tested scenarios.

## 6.5 Categorical Autoencoder

Table 11 lists each categorical feature together with its cardinality and per-feature reconstruction accuracy. The results indicate perfect or near-perfect reconstruction for almost every variable. The most complex feature, ToothType, with a cardinality of 32, is decoded correctly 95% of the time, whereas all remaining features exceed 99.9%.

Table 11: Per-feature reconstruction accuracy of the CAE

| **Feature** | **Cardinality** | **Accuracy** |
|---|---|---|
| ToothType | 32 | 0.9511 |
| TumorLaterality | 2 | 1.0000 |
| TumorLocation | 8 | 0.9994 |
| TNM_T | 5 | 1.0000 |
| TNM_N | 4 | 0.9994 |

| **Feature** | **Cardinality** | **Accuracy** |
|---|---|---|
| TNM_Stage | 4 | 1.0000 |
| TumorCategory | 6 | 1.0000 |
| TumorSite | 9 | 1.0000 |
| Stage | 7 | 1.0000 |
| SystemicTherapy | 5 | 1.0000 |
| StageItmIV | 5 | 1.0000 |

Table 20 in Appendix I compares the evaluation metrics of the baseline SVMs with rbf kernel with a version that reduced OHE features extracted from the AE. Across all evaluation metrics the baseline model slightly outperforms the AE-enhanced model, improving BS and accuracy by roughly 0.02 in both settings. The most reasonable explanation for this minor difference is the imperfect reconstruction of the high-cardinality ToothType variable. The performance gap remains small,indicating that the latent representation of the categorical AE still preserves most of the original predictive signal.

# 7 Conclusion

To conclude, this study is set out to evaluate whether ML can assist in guiding pre-RT tooth extraction decisions for patients with HNC using limited clinical features. Accordingly, it assessed the predictive performance of conventional classifiers, analyzed feature importance, and addressed the challenges posed by high-cardinality categorical data. Specifically, categorical kernels within SVMs were used to eliminate the need for OHE, with additional attempts to improve the categorical kernel proposed by Belanche and Villegas (2013). The study further explored CAE for dimensionality reduction and integrated them into SVM models. Finally, this study provides is the first to integrate of categorical kernels into ensemble SVMs. Based on this work, the study's findings enable explicit answers to the RQs.

- **RQ1**: Can ML models reliably predict tooth extraction necessity using limited pre-RT clinical data?

Across both $D_{mean}$ and $D_{max}$-based predictions, LR consistently underperformed relative to SVM and RF. The SVM and RF demonstrated a better balance between recall and precision, with stronger calibration and reliable performance in high-confidence regions. All models exhibited uncertainty in intermediate probability ranges for both datasets. However, a key limitation for clinical implementation is that further validation is required to establish reliable probability thresholds that could automatically guide extraction decisions versus those requiring clinical review.

- **RQ2:** Which features are most important in the models for accurately classifying the need for tooth extraction?

The analysis shows that, across all models and for both dose metrics, the most influential predictors are the tumor's location and the type of the tooth in question. In contrast, variables such as staging, prescribed dose, and treatment factors play only a secondary role.

- **RQ3:** Does the proposed extension of the categorical kernel by Belanche and Villegas (2013) improve performance over their original kernel?

The adjusted kernel $k_2$ converged to hyperparameters that effectively reduced it to the original kernel $k_1$. Consequently, $k_1$ and $k_2$ performed similarly. Hence, allocating nonzero similarity to unrelated categories provided no benefit.

- **RQ4:** Can a CAE provide a more effective alternative to OHE for training SVMs on high-dimensional categorical medical data?

On the tooth extraction dataset, the CAE reconstructed near-perfect inputs. It only had a minor errors on the highest-cardinality variable, namely, ToothType. The SVMs trained on the CAE embeddings matched the performance of OHE-based SVMs. However, the datasets did not include extremely high cardinalities. Therefore, testing CAEs on features with hundreds or thousands of categories prior to modelling is necessary.

- **RQ5:** Do ensemble SVMs with categorical kernels outperform ensembles using standard kernels or a single SVM?

RSM and RSVM ensembles using the categorical kernel $k_1$ underperformed compared with ensembles using rbf or linear kernels. In addition, it did not surpass a well-tuned single SVM with $k_1$, rbf, or linear kernel. Varying the ensemble size had little impact on accuracy. Including the entire feature set was crucial to the performance. For instance, $k_1$ ensembles significantly suffered when features were subsampled. Taken together, these results do not support using $k_1$ within the RSM or RSVM models for this task.

Future work could explore several directions to extend and strengthen these findings. First, rather than modeling the binary risk of exceeding a fixed 40 Gy threshold, deep learning approaches trained on historical RT plans could be used to directly predict the continuous dose received by each tooth. Such models may offer more clinically informative guidance. Second, in the context of categorical kernels, it would be valuable to move beyond Hamming-distance formulations and investigate alternative kernel constructions. Finally, for AE-based approaches, future studies could experiment with deeper architectures and more expressive functions, particularly on features with very high cardinality, to assess whether reconstruction quality and predictive performance can be improved beyond what was achieved with the current CAE design.

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# Appendix A Sustainable Development Goals (SDG) Statement

| | |
|---|---|
| Name: | Asha Sharon de Meij |
| ID: | i6254733 |
| Supervisor: | Rui Jorge de Almeida |
| Date: | November 1, 2025 |

Through the research conducted for this master's thesis, I seek to contribute to one or more of the 17 SDG(s) set forth by the United Nations (https://www.undp.org/sustainable-development-goals).


SDG Codes: SDG 3 (Good Health and Well-being) and SDG 9 (Industry, Innovation and Infrastructure).

This thesis aligns most directly with the United Nations Sustainable Development third goal. The central focus of this research is to improve the quality of care for patients with HNC by supporting clinicians in making more informed pre-RT tooth extraction decisions. By introducing ML models that more accurately distinguish which teeth truly require removal, this research provides a means to minimize avoidable procedures and preserve oral function. Improved decision support contributes to patient well-being by supporting better treatment adherence and recovery outcomes. Preserving teeth where safely possible helps maintain nutrition and body weight, both of which are crucial for completing RT as planned and achieving optimal recovery.

At the same time, the research also resonates with the ninth goal. Applying ML to guide pre-RT extraction decisions represents an innovative approach in oncology, as such tools are not yet established in routine clinical practice. Furthermore, by developing and evaluating new methodological strategies, such as categorical kernels, AEs, and ensemble frameworks, this thesis demonstrates how computational innovation can be utilized to improve healthcare delivery. In this sense, the work not only advances patient outcomes but also contributes to building innovative infrastructures in medical decision support, bridging the gap between methodological research and clinical application.

# Appendix B World Dental Federation Tooth Numbering System

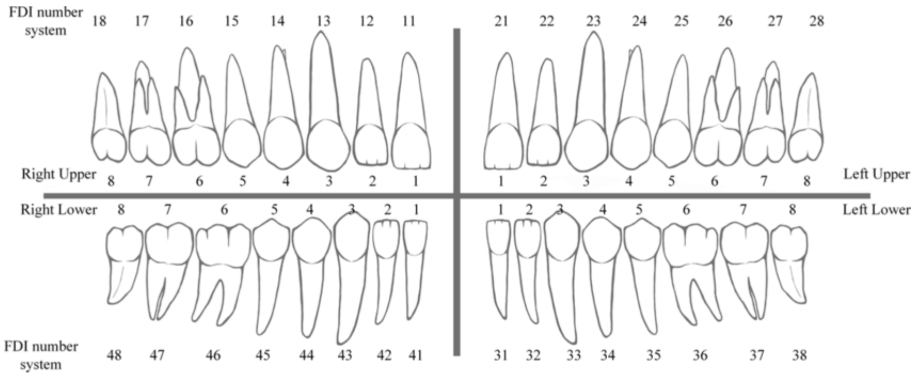


Figure 19: Schematic diagram of the World Dental Federation two-digit tooth numbering system. (Lee et al. 2019)

# Appendix C Random Subspace Support Vector Machine Algorithm

**Algorithm 1** Random Subspace Models for SVMs (Bos 2024)

**Input:**

- Training set $\mathcal{D}_{\text{train}} = \{(X_i, y_i) \mid i = 1, \dots, N\}$ with $X_i = (X_{i1}, \dots, X_{ip})^T \in \mathbb{R}^p$ and $y_i \in \mathcal{L} = \{1, \dots, K\}$
- Number of iterations: $B \in \mathbb{N}_+$
- Number of features to use in each iteration: $m$
- Number of folds in cross-validation: $k \in \mathbb{N}_+$
- Kernel type (linear, polynomial, radial basis, etc.): $\mathcal{K}$
- Parameter grid $\mathcal{G}$

**Output:** An ensemble list of $B$ models and their cross-validation estimated accuracy

Initialize a list collecting the ensemble: $E$

Initialize a list collecting the estimated accuracies of the models in the ensemble: $\mathcal{A}$

**for** $b = 1$ to $B$ **do**

  Draw a random set of features $\mathcal{F}_b = \{f_j \mid j = 1, \dots, m\} \subset \{1, \dots, p\}$ of size $m$

  Create a sub-dataset $\mathcal{D}_{\text{train}}^{\mathcal{F}_b}$ using only features in $\mathcal{F}_b$, but all $N$ observations, i.e.

$$\mathcal{D}_{\text{train}}^{\mathcal{F}_b} = \{(X_i^{\mathcal{F}_b}, y_i) \mid X_i^{\mathcal{F}_b} \in \mathbb{R}^m,\ y_i \in \mathcal{L},\ i = 1, \dots, N\}$$

  where $X_i^{\mathcal{F}_b} = (X_{if_1}, X_{if_2}, \dots, X_{if_m})^T$

  Use cross validation to find the tuning parameters:

  - $results \leftarrow \text{CrossValidation}(\mathcal{D}_{\text{train}}^{\mathcal{F}_b}, k, \mathcal{K}, \mathcal{G})$
  - Find the parameter set $TP_b^*$ with the highest accuracy in *results*
  - Store this highest accuracy in $\mathcal{A}_b$

  $\mathcal{A} \leftarrow \mathcal{A} \cup \mathcal{A}_b$

  $optim\_model \leftarrow \text{Train SVM}(\text{kernel} = \mathcal{K}, \text{parameters} = TP_b^*, \text{data} = \mathcal{D}_{\text{train}}^{\mathcal{F}_b})$

  $E \leftarrow E \cup optim\_model$

**end for**

**return** $(E, \mathcal{A})$

# Appendix D Random Support Vector Machine Algorithm

**Algorithm 2** SubBag model for SVM (Bos 2024)

**Input:**

- Training set $\mathcal{D}_{\text{train}} = \{(X_i, y_i) \mid i = 1, \ldots, N\}$ with $X_i = (X_{i1}, \ldots, X_{ip})^T \in \mathbb{R}^p$ and $y_i \in \mathcal{L} = \{l_1, \ldots, l_K\}$
- Number of iterations: $B \in \mathbb{N}_+$
- Number of features to use in each iteration: $m$
- Number of folds in cross-validation: $k \in \mathbb{N}_+$
- Kernel type (linear, polynomial, radial basis, etc.): $\mathcal{K}$
- Parameter grid $\mathcal{G}$

**Output:** An ensemble list of $B$ models and their cross-validation estimated accuracy

Initialize a list collecting the ensemble: $E$
Initialize a list collecting the estimated accuracies of the models in the ensemble: $\mathcal{A}$
**for** $b = 1$ to $B$ **do**
    Draw a random set of features $\mathcal{F}_b = \{f_j^b \mid j = 1, \ldots, m\} \subset \{1, \ldots, p\}$
    Draw a bootstrap sample of indices $I_b = \{i_1, i_2, \ldots, i_N\} \subset \{1, \ldots, N\}$ (with replacement)
    Create a sub-dataset using only features in $\mathcal{F}_b$ and observations in $I_b$:

$$\mathcal{D}_{\text{train}}^{(b)} = \left\{(X_i^{\mathcal{F}_b}, y_i) \mid i \in I_b\right\}, \quad \text{where } X_i^{\mathcal{F}_b} = (X_{if_1^b}, X_{if_2^b}, \ldots, X_{if_m^b})^T$$

    Use cross-validation to find the tuning parameters:
    - $results \leftarrow \text{CrossValidation}(\mathcal{D}_{\text{train}}^{(b)}, k, \mathcal{K}, \mathcal{G})$
    - Find the parameter set $TP_b^*$ with the highest accuracy in *results*
    - Store this highest accuracy in $\mathcal{A}_b$

    $\mathcal{A} \leftarrow \mathcal{A} \cup \{\mathcal{A}_b\}$
    $optim\_model \leftarrow \text{TrainSVM}(= \mathcal{K}, = TP_b^*, = \mathcal{D}_{\text{train}}^{(b)})$
    $E \leftarrow E \cup \{optim\_model\}$
**end for**
**return** $(E, \mathcal{A})$

# Appendix E Hyperparameter Grids of Models

For the models presented in Tables 12 - 15 GridSearchCV with 5-fold CV was employed.

Table 12: LR Hyperparameter Grid

| **Hyperparameter** | **Values** |
|---|---|
| C | {0.01, 0.1, 1, 10} |
| class_weight | {balanced, {0:1,1:2}, {0:1,1:4}} |

Table 13: SVM Hyperparameter Grid

| **Hyperparameter** | **Values** |
|---|---|
| kernel | `linear, rbf` |
| C | {0.001, 0.01, 0.1, 1, 10, 100} |
| class_weight | {balanced, {0:1,1:2}, {0:1,1:4}} |

Table 14: RF Hyperparameter Grid

| **Hyperparameter** | **Values** |
|---|---|
| n_estimators | {100, 200, 300, 500} |
| max_depth | {10, 20, 30} |
| min_samples_split | {2, 5, 10} |
| min_samples_leaf | {1, 2, 4} |
| class_weight | {balanced, {0:1,1:2}, {0:1,1:4}} |

Table 15: Categorical Kernel SVM Hyperparameter Grid

| **Kernel** | **C** | **alpha** | **gamma** | **sigma** |
|---|---|---|---|---|
| $k_1$ | {0.1, 1, 10} | {0.1, 0.2, 0.3, 0.5, 0.7, 0.9, 1.0, 1.5} | {0.125, 0.25, 0.5, 1, 2, 4} | None |
| $k_2$ | {0.1, 1, 10} | {0.1, 0.2, 0.3, 0.5, 0.7, 0.9, 1.0, 1.5} | {0.125, 0.25, 0.5, 1, 2, 4} | {0.001, 0.01, 0.05, 0.1, 0.125, 0.2, 0.5} |

For the ensemble SVM models a 3-fold CV is used to reduce computational cost given the larger number of base learners and parameter combinations. Furthermore the number of hyperparameters was also reduced. Ensembles are constructed using three types of kernels: the categorical kernel $k_1$, as well as standard rbf and linear kernels. The grid is shown in Table 16.

Table 16: Ensemble SVM Hyperparameter Grid

| **Kernel** | **C** | **alpha** | **gamma** |
|---|---|---|---|
| `linear` | {0.1, 1, 10} | None | None |
| `rbf` | {0.1, 1, 10} | None | None |
| $k_1$ | {0.1, 1, 10} | {0.3, 0.5, 0.7, 1.0, 1.5} | {0.5, 1.0, 2.0, 4.0} |

Table 17 summarizes the hyperparameters and functions employed in the CAE used in this study.

Table 17: CAE Model setup

| **Component** | **Specification** |
|---|---|
| Input dimension | Total number of OHE variables (m) |
| Bottleneck size | $max(2, \lfloor m/2 \rfloor))$ |
| Encoder | Dense layer, linear activation, no bias |
| Decoder | Dense layer with bias + per-feature softmax heads |
| Optimizer | Adam, learning rate 0.01 |
| Loss | Categorical cross-entropy |
| Batch size | 64 |
| Epochs | Up to 50 (EarlyStopping on validation loss, patience = 5) |
| Validation split | 10% |

# Appendix F Demonstration of Categorical Kernels $k_1$ and $k_2$

This section provides a concrete example to illustrate how the $k_1$ and $k_2$ kernels work and demonstrate the key differences between them. The following dataset with six instances and three categorical features serves to compute the kernel matrices and to demonstrate the difference in the resulting similarity structure:

| | Color | Brand | Fuel Type |
|---|---|---|---|
| $r_1$ | Red | Toyota | Hybrid |
| $r_2$ | Red | BMW | Gasoline |
| $r_3$ | Blue | BMW | Diesel |
| $r_4$ | Black | Mercedes | Hybrid |
| $r_5$ | Black | Toyota | Gasoline |
| $r_6$ | Green | Ford | Electric |

The computation of a univariate kernel is demonstrated using the 'Color' column. First, the empirical probabilities are calculated. Then, applying the transformation $h_\alpha$ with $\alpha = 1.5$ gives:

$$P(\text{Red}) = 0.333 \Rightarrow h_\alpha(0.333) \approx 0.867$$

$$P(\text{Blue}) = 0.167 \Rightarrow h_\alpha(0.167) \approx 0.953$$

$$P(\text{Black}) = 0.333 \Rightarrow h_\alpha(0.333) \approx 0.867$$

$$P(\text{Green}) = 0.167 \Rightarrow h_\alpha(0.167) \approx 0.953$$

The $k_1$ kernel assigns a positive similarity only to instances that share the exact same category level. Whereas, different levels are considered completely dissimilar. For example, the univariate kernel matrix, $K^{(U)}_{1,F1}$, for the 'Color' feature is:

$$K^{(U)}_{1,\,F1} = \begin{bmatrix} 0.867 & 0.867 & 0 & 0 & 0 & 0 \\ 0.867 & 0.867 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.953 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.867 & 0.867 & 0 \\ 0 & 0 & 0 & 0.867 & 0.867 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.953 \end{bmatrix}$$

In contrast, the $k_2$ kernel allows different levels to be considered as similar by assigning scores to them. For instance, while ’Red’ and ’Black’ are different categories, they share the same empirical probability (0.333), leading to a high similarity score:

$$k_2(\text{Red, Black}) = \exp\left(-\frac{(0.333-0.333)^2}{0.1^2}\right)\sqrt{0.867\times 0.867} = 0.867.$$

Conversely, ’Blue’ and ’Green’, both being rare, also receive a high score:

$$k_2(\text{Blue, Green}) = \exp\left(-\frac{(0.167-0.167)^2}{0.1^2}\right)\sqrt{0.953\times 0.953} = 0.953.$$

However, pairs with a significant probability difference, like ’Red’ and ’Blue’, are considered highly dissimilar:

$$k_2(\text{Red, Blue}) = \exp\left(-\frac{(0.333-0.167)^2}{0.01}\right)\sqrt{0.867\times 0.953} \approx 0.058.$$

In the $k_2$ kernel, a larger $\sigma$ reduces the penalty for probability differences, resulting in higher off-diagonal similarities, while a smaller $\sigma$ strengthens the penalty, making the kernel more sensitive to differences in empirical rarity. The resulting univariate kernel matrix for the ’Color’ feature, $K^{(U)}_{2,F_1}$, is therefore:

$$K_{2,F1}^{(U)} = \begin{bmatrix} 0.867 & 0.867 & 0.0577 & 0.867 & 0.867 & 0.0577 \\ 0.867 & 0.867 & 0.0577 & 0.867 & 0.867 & 0.0577 \\ 0.0577 & 0.0577 & 0.953 & 0.0577 & 0.0577 & 0.953 \\ 0.867 & 0.867 & 0.0577 & 0.867 & 0.867 & 0.0577 \\ 0.867 & 0.867 & 0.0577 & 0.867 & 0.867 & 0.0577 \\ 0.0577 & 0.0577 & 0.953 & 0.0577 & 0.0577 & 0.953 \end{bmatrix}$$

Having demonstrated the calculation for the 'Color' feature, the same process is applied to compute univariate $k_1$ and $k_2$ kernels for the other categorical features, 'Brand' and 'Fuel Type'. These individual univariate kernels are then aggregated into a final multivariate kernel using the exponential formulation:

$$k_n(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(\frac{\gamma}{d}\sum_{k=1}^{d} k_n^{(U)}(x_{ik}, x_{jk})\right)$$

Using the parameters $\alpha = 1.5$, $\sigma = 0.1$, $\gamma = 1.0$, and $d = 3$ (the number of features), the resulting full categorical kernel matrices are as follows.

For the $k_1$ kernel:

$$K_{\text{cat}}^{(k_1)} = \begin{bmatrix} 2.380 & 1.335 & 1.000 & 1.335 & 1.335 & 1.000 \\ 1.335 & 2.380 & 1.335 & 1.000 & 1.335 & 1.000 \\ 1.000 & 1.335 & 2.519 & 1.000 & 1.000 & 1.000 \\ 1.335 & 1.000 & 1.000 & 2.449 & 1.335 & 1.000 \\ 1.335 & 1.335 & 1.000 & 1.335 & 2.380 & 1.000 \\ 1.000 & 1.000 & 1.000 & 1.000 & 1.000 & 2.594 \end{bmatrix}$$

For the $k_2$ kernel:

$$K_{\text{cat}}^{(k_2)} = \begin{bmatrix} 2.380 & 1.387 & 1.040 & 1.387 & 1.387 & 1.040 \\ 1.387 & 2.380 & 1.040 & 1.387 & 1.387 & 1.040 \\ 1.040 & 1.040 & 2.594 & 1.040 & 1.040 & 2.594 \\ 1.387 & 1.387 & 1.040 & 2.380 & 1.387 & 1.040 \\ 1.387 & 1.387 & 1.040 & 1.387 & 2.380 & 1.040 \\ 1.040 & 1.040 & 2.594 & 1.040 & 1.040 & 2.594 \end{bmatrix}$$

# Appendix G Categorical Kernel Results

Table 18: Performance of SVM models on MONK's Problems dataset

| **Kernel** | **Acc** | **Class 0** | | | **Class 1** | | |
|---|---|---|---|---|---|---|---|
| | | P | R | $F_1$ | P | R | $F_1$ |
| rbf | 1.0000 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| $k_1$ | 1.0000 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| $k_2$ | 1.0000 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |

Table 19: Performance of SVM models on Congressional Voting Records dataset

| **Kernel** | **Acc** | **Class 0** | | | **Class 1** | | |
|---|---|---|---|---|---|---|---|
| | | P | R | $F_1$ | P | R | $F_1$ |
| rbf | 0.9500 | 0.94 | 0.94 | 0.94 | 0.96 | 0.96 | 0.96 |
| $k_1$ | 0.9700 | 0.94 | 0.97 | 0.95 | 0.98 | 0.96 | 0.97 |
| $k_2$ | 0.9500 | 0.94 | 0.94 | 0.94 | 0.96 | 0.96 | 0.96 |

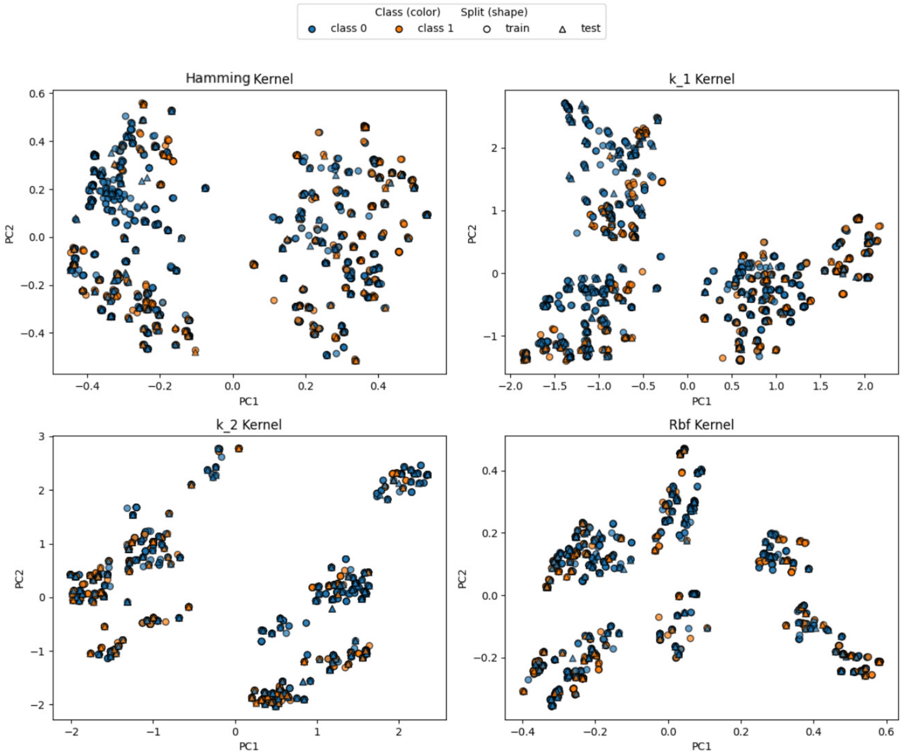


Figure 20: Data projection of $D_{mean}$ dataset via KernelPCA using Hamming, $k_1$, $k_2$, and rbf kernels

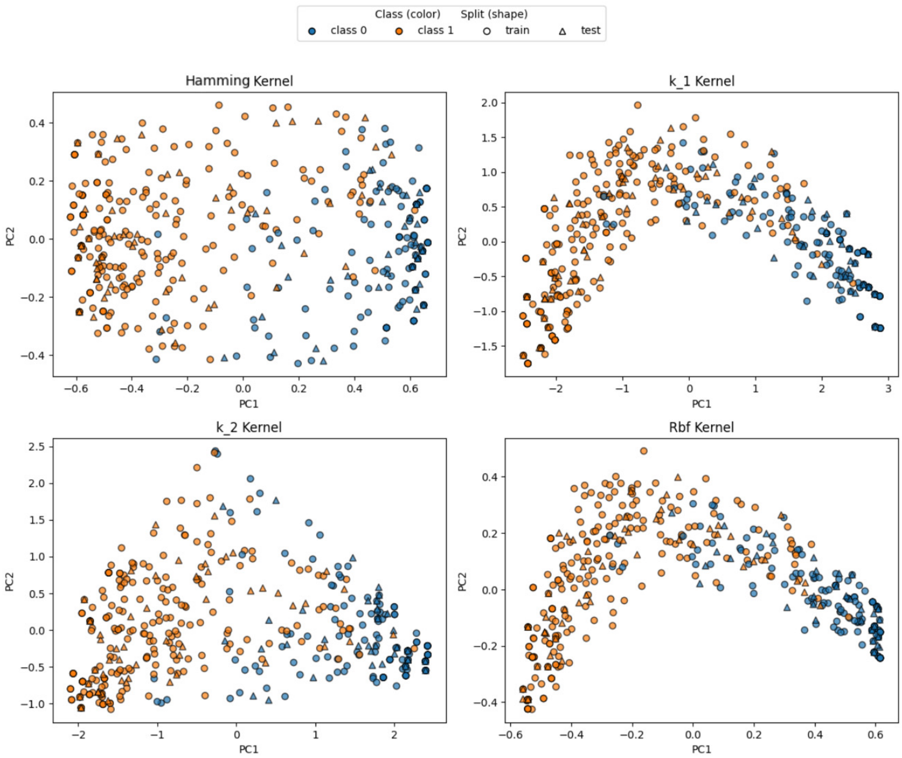


Figure 21: Data projection of Congressional Vote dataset via KernelPCA using Hamming, $k_1$, $k_2$, and rbf kernels

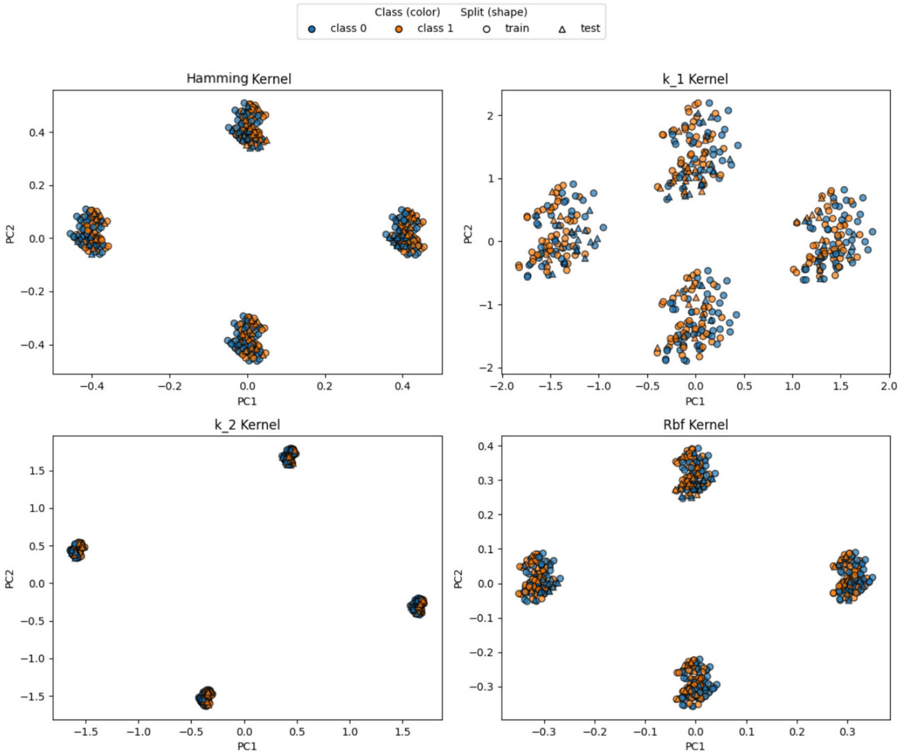


Figure 22: Data projection of MONK's Problems dataset via KernelPCA using Hamming, $k_1$, $k_2$, and rbf kernels

# Appendix H Ensemble Support Vector Machine Results

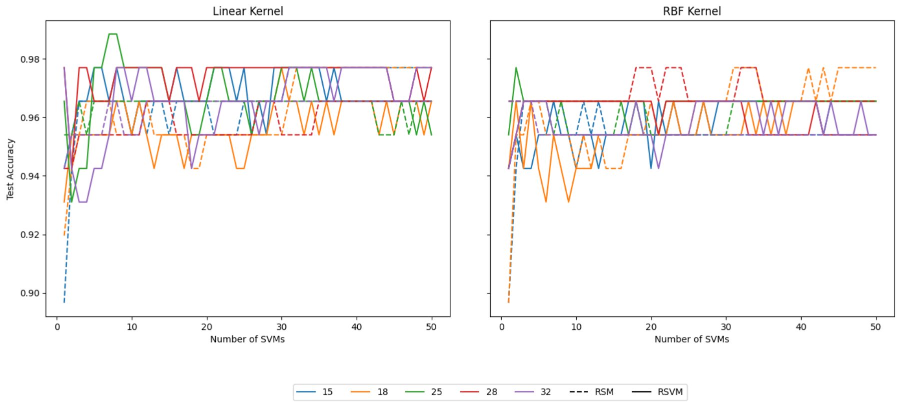


Figure 23: Performance of SVM ensembles with linear and rbf kernels across different numbers of features ($m$) and ensemble sizes on Congressional Voting dataset

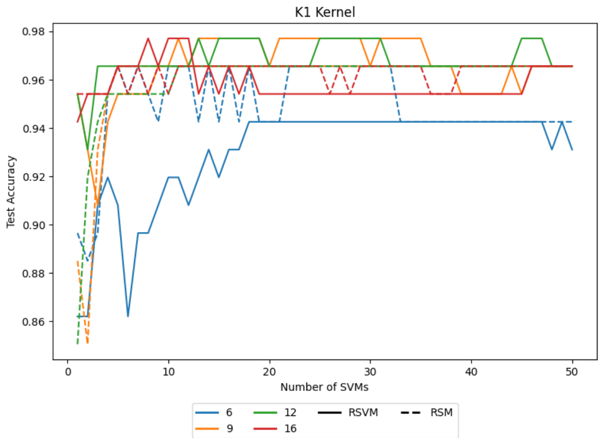


Figure 24: Performance of SVM ensemble with $k_1$ kernel across different numbers of features ($m$) and ensemble sizes on Congressional Voting dataset

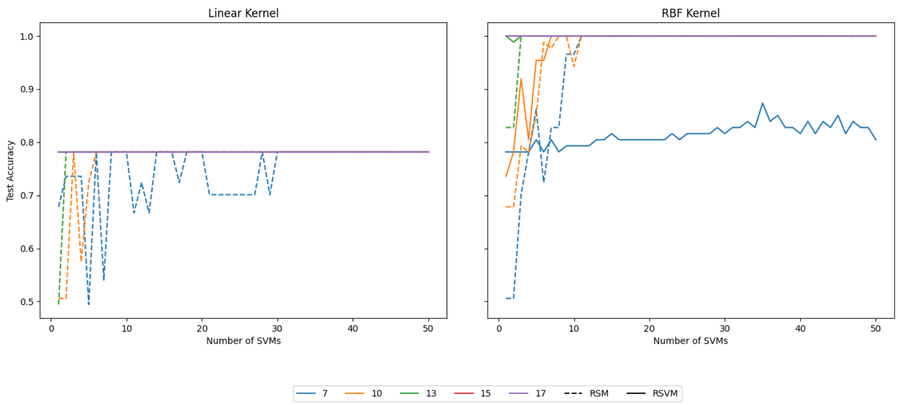


Figure 25: Performance of SVM ensembles with linear and rbf kernels across different numbers of features ($m$) and ensemble sizes on MONK's Problems dataset

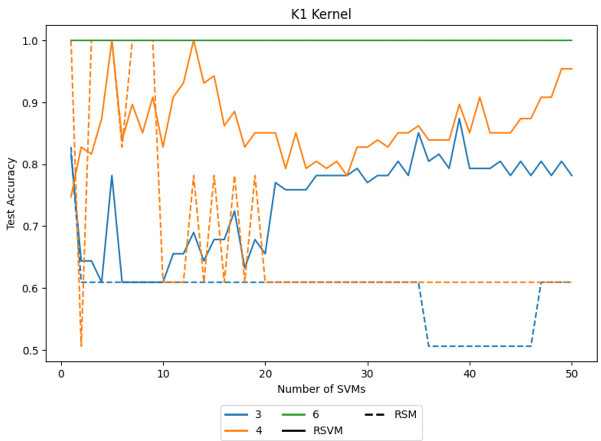


Figure 26: Performance of SVM ensemble with $k_1$ kernel across different numbers of features ($m$) and ensemble sizes on the MONK's Problems dataset

# Appendix I Support Vector Machine Results with OHE and Compressed Features

Table 20: Performance of SVM with a rbf kernel using OHE vs. CAE-compressed features on the $D_{mean}$ and $D_{max}$ datasets

| **Dataset** | **Features** | **BS** | **bCE** | **Acc** | **Class 0** | | | **Class 1** | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | P | R | $F_1$ | P | R | $F_1$ |
| $D_{mean}$ | OHE | 0.0971 | 0.4257 | 0.87 | 0.92 | 0.91 | 0.91 | 0.72 | 0.73 | 0.73 |
| | AE features | 0.1203 | 0.4827 | 0.85 | 0.88 | 0.91 | 0.90 | 0.76 | 0.70 | 0.73 |
| $D_{max}$ | OHE | 0.1156 | 0.3823 | 0.84 | 0.85 | 0.86 | 0.86 | 0.82 | 0.80 | 0.81 |
| | AE features | 0.1365 | 0.4446 | 0.82 | 0.85 | 0.85 | 0.85 | 0.78 | 0.78 | 0.78 |

# Appendix J Declaration of Originality MSc Thesis

By signing this statement, I hereby acknowledge the submitted MSc Thesis titled

Machine Learning for Pre-Radiotherapy Tooth Extraction Prediction

to be produced independently by me, without external help. Wherever I paraphrase or cite literally, a reference to the original source (journal, book, report, internet, etc.) is provided. By signing this statement, I explicitly declare that I am aware of the fraud sanctions as stated in the Education and Examination Regulations (EERs) of SBE, Maastricht University.

| | |
|---|---|
| Place: | Maastricht |
| Date: | November 7, 2025 |
| Study programme: | MSc Econometrics and Operations Research |
| Signature: | Asha Sharon de Meij |